



भारत सरकार / Government of India
विद्युत मंत्रालय / Ministry of Power
पूर्वी क्षेत्रीय विद्युत समिति / Eastern Regional Power Committee

सं /NO. ERPC/EE/OPERATION/2026/ 1073

DATE:28.08.2026

सेवा में /To
संलग्न सूची के अनुसार /As per list enclosed.

विषय : 20.08.2026 (गुरुवार) को ईआरपीसी सचिवालय, कोलकाता में भौतिक रूप से आयोजित 242वीं ओसीसी बैठक के कार्यवृत्त - के संबंध में

Sub: Minutes of 242nd OCC Meeting held on 20.08.2026 (Thursday) physically at ERPC Secretariat, Kolkata – reg

महोदय/महोदया,
Sir(s)/Madam,

कृपया अपनी जानकारी और आवश्यक कार्रवाई के लिए 20 अगस्त 2026 (गुरुवार) को आयोजित 242वीं ओसीसी बैठक के कार्यवृत्त संलग्न देखें। यह ईआरपीसी वेबसाइट (www.erpc.gov.in) पर भी उपलब्ध है।

Please find enclosed **Minutes of 242nd OCC Meeting** held on 20.08.2026 (Friday) physically at ERPC Secretariat, Kolkata at 10:30 hrs. This is for your kind information and necessary action. The same is also available at ERPC website (www.erpc.gov.in).

टिप्पणियाँ, यदि कोई हों, कृपया यथाशीघ्र इस कार्यालय को अग्रेषित करें।
Observations, if any, may please be forwarded to this office at the earliest.

इसे सदस्य सचिव के अनुमोदन से जारी किया जाता है।
This issues with the approval of Member Secretary.

भवदीय /Yours faithfully


(B.S.Ray)

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Eastern Regional Power Committee

MINUTES OF 242nd OCC MEETING

Venue: ERPC Secretariat, Kolkata

Date: 20.08.2026

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EASTERN REGIONAL POWER COMMITTEE

MINUTES OF 242nd OCC MEETING HELD ON 20.08.2026 (THURSDAY) AT 10:30 HRS

Member Secretary, ERPC welcomed all OCC members and other participants who were present physically as well as those who had joined virtually to the **242nd OCC Meeting**.

Thereafter he requested all the participants to give brief introduction.

□ He outlined the performance of ER grid during **July-2026** highlighting the following points:

- In **July-2026**, energy consumption of ER was **20727 MU** which is **6.4 % more** than **July -2025**.
- In **July-2026**, Peak demand met of ER was **32391 MW** which is **0.3 % more** than **July -2025**.
- ❖ Thermal **PLF** of **ER** during **July-2026** was **75%** whereas National average is **68.2%**.
- ❖ Generating stations whose PLF was **90%** or more during **July-2026**:

Utility	Generating station	PLF (%)
WBPDC	BAKRESWAR TPS	91
IPP	HALDIA TPP (HEL)	99

- ❖ During **July-2026**, **72.5 %** of the time, the ER grid frequency was within IEGC Band (49.90Hz-50.05Hz).

Coal stock position (As on 17.08.2026) is as follows:

SL.	Name of States/Power Stns.	% of Actual Stock vis-à-vis Normative Stock
1.	Jharkhand (TVNL)	95%
2.	Odisha (IB TPS)	100%
3.	West Bengal (WBPDC)	69%
4.	D.P.L. TPS	33%
5.	DVC	90%
6.	NTPC	62%

Coal stock position was reported satisfactory across constituents except one utility (DPL), where stock levels were flagged as depleted.

- It was informed that the National Power Committee (NPC) meeting is scheduled to be held on 18th September 2026; agenda items proposed by ER constituents (if any) for that meeting may be forwarded at the earliest.

ED, ERLDC touched upon the following key facets:

- The prevailing pattern of ER demand was highlighted, especially recent low demand being witnessed owing to mild weather conditions with heavy rain but demand is expected to surge in upcoming months with festival season.

- *Significant concerns were raised regarding multiple simultaneous line-shutdown requests linked to National Highways Authority of India (NHAI) four-lane highway construction works crossing transmission corridors at various locations, which poses a threat to grid reliability if not properly coordinated and staggered.*
- *Onset of generation from all 3 units of TEESTA V HPS (NHPC) was highlighted. NHPC was appreciated for this development.*

*List of participants attached at **Annex-A**.*

Member Secretary, ERPC requested S.E (Operation) to take up the agenda for discussion.

1. PART-A: CONFIRMATION OF MINUTES

1.1. Confirmation of Minutes of 241st OCC Meeting held at ERPC Secretariat on 17th July 2026

The Minutes of 241st Operation Coordination Sub-Committee meeting held on 17.07.2026 was circulated vide letter dated 24.07.2026.

Deliberations in the meeting

In the MOM of 241st OCC meeting held on 17.07.2026, certain portions of **Item No: B.2.18** was requested to modify from PGCIL as detailed below:

ORIGINAL	REVISED
"Powergrid submitted: ➤ On account of space constraints at the existing base/substation, a pedestal-mounted wave trap arrangement is being proposed in place of the existing overhead/hanging wave-trap arrangement, which currently lacks a dedicated support structure at the 220 kV bay. ➤ The equipment presently available in the market corresponds to the hanging-type design, and the specification would need to be revisited to suit the pedestal-mounted requirement. ➤ The proposed (pedestal-mounted) arrangement is cost-effective compared to any other available alternative. OCC Decision ✓OCC decided that Powergrid shall finalize the design/specification for the pedestal-mounted wave trap in consultation with CTU. Accordingly, the same may be placed in CMETS-ER for concurrence. ✓Cost-sharing aspects also need to be examined by Powergrid and to be placed in TCC/ERPC forum for approval. "	" 1. POWERGRID proposed to change the scope already issued by CTU, w.r.t problem faced for placing conventional Wave Trap in old stations of Chukha TS due to space constraint. 2. In existing CTS SS, hanging type Wave Trap design is available, however due to reconductoring with HTLS and subsequent enhancement of weight in accordance with higher current rating, hanging type Wave Trap is not feasible for existing gantry tower structures and further, if Wave Trap (With higher rating) required to be installed, then, pedestal mounted conventional design to be adopted. To accommodate pedestal in existing SS, space constraint is observed as many other line side items also required to be shifted which is not possible for existing SS. 3. To avoid the criticality, feasible option is to install DTPC for both channels, and in that process, there will be no requirement of wave trap installations and above criticality also will be avoided, completely. OCC Decision ✓OCC decided that Powergrid shall finalize the design/specification for the wave trap in consultation with CTU. Accordingly, the same may be placed in CMETS-ER for concurrence. ✓Cost-sharing aspects also need to be examined by Powergrid and to be placed in TCC/ERPC forum for approval."

Forum agreed with the modification and confirmed the Minutes of 241st OCC with above mentioned modification.

2. PART-B: ITEMS FOR DISCUSSION

2.1 Update on follow up agenda: ERPC

a) Intrastate Transmission Network Assessment & Mitigation-Odisha

Reference:

Implementation of the Under Voltage Load Shedding (UVLS) scheme in the Odisha system has been under review since the 231st, 232nd, 233rd, and 234th OCC Meetings held on 22.09.2025, 24.10.2025, 22.11.2025, and 23.12.2025 respectively.

The matter was also discussed in the recently concluded 55th TCC/ERPC meeting held on 16.12.2025 and 17.12.2025 at Kalimpong, West Bengal. As per deliberation in 55th TCC Meeting, **300MW load** has been identified and it will be operationalized before **Summer '26**.

The continued delay in implementation is posing increasing risks not only to the Odisha system but also to the Eastern Region as a whole during the forthcoming Summer-2026 period.

- During a review meeting held on 30.04.2026, OPTCL and SLDC Odisha informed that the quotation from the existing ADMS vendor had been received and was under examination. Further, 31 feeders of TPCODL and 23 feeders of TPNODL have been identified under the scheme. The target date for completion was indicated as June-2026.

56th ERPC Decision:

ERPC forum advised OPTCL to implement the UVLS scheme by June-2026 without any further delay. Detailed work plan for reconductoring of intrastate transmission line shall be shared with ERPC & ERLDC within two weeks.

As per 241st OCC:

OPTCL informed:

- The UVLS scheme is being implemented through ADMS in two phases (200 MW each).
- Thirty of the thirty-five Grid Substations are already covered under ADMS while the remaining substations are under integration.
- The agency is carrying out logic testing and installation of RTU components.

241st OCC Decision

- ✓ In view of system security, OCC stressed the importance of completing implementation of at least Phase-I of ADMS on priority basis.
- ✓ OPTCL shall implement the first phase (200 MW) of UVLS without further delay and complete the balance work by August 2026. Weekly progress reports along with a detailed implementation timeline i.r.o individual feeders shall be submitted to ERPC.

Deliberations in the meeting

OPTCL informed:

- *The scheme is being implemented through the Automatic Demand Management System (ADMS) in two phases of 200 MW each;*

- The executing agency has finalised the software logic and functional code, which was submitted to ERLDC on 10th August'2026 for confirmation before final implementation.
- Out of 35 grid sub-stations selected for the scheme, 30 already have ADMS controllers installed; the remaining 5 substations is under progress. Wiring has been completed at 12 of the 35 grid stations so far.
- All wiring works would be completed within approximately 45 days starting from the OCC date.

SLDC Odisha informed that the software logic considered by them for integration of UVLS through ADMS had been shared with ERLDC on 19.08.2026 seeking opinion of ERLDC on the same.

OCC Decision

- OPTCL was advised to complete UVLS implementation by 20th September 2026. Weekly progress reports to be submitted to ERPC/ERLDC.
- OCC also consented to the proposed logic and functional flow of the UVLS logic as submitted by OPTCL (**Annex B.2.1.(a)**).

b) Urgent Review of Phase I Compliance for CEA Flexible Operation Regulations, 2023

- ☐ CEA has issued a gazette notification dated January 30, 2023, regarding flexible operation of coal-fired generating units.
- ☐ As per CEA gazette notification extraordinary, part III, section 4, no. 61 (CG-DL-E-31012023-243299), the coal-based power generating units shall have flexible operation capability with minimum power level of 55%. This capability includes a ramp rate of 2% between 55%-70% and a ramp rate of 3% above 70%. This capability was mandated within one year of the notification, i.e., by Jan 2024. Further, the regulation mandates that generating units not capable of achieving a minimum power level of 40% shall achieve the same as per the CEA phasing plan.
- ☐ It may be noted that CEA's flexible operation regulations were notified after successful pilot test/study conducted across the country in association with international partners and BHEL.
- ☐ In this regard, it is pertinent to mention that 91 Units of total installed Capacity 51,080 MW in aggregate of various thermal power plants have been notified under Phase I (July 2024 – June 2026) for operation at 40% MTL (Minimum Technical Load). These units are required to carry out the necessary retrofitting/modifications as recommended by the respective OEMs so as to be fully prepared for sustained operation at 40% MTL by June 2026, as mandated under the provisions of the aforesaid Regulations.
- ☐ It may also be noted that successful and sustained operation at 40% MTL with Indian coal has been demonstrated by other generating utilities (Under Pilot Phase) such as West Bengal Power Development Corporation Limited (WBPDCCL) at Sagardighi Unit-8 and Damodar Valley Corporation (DVC) at Mejia Unit-8, following suitable retrofits as advised by the OEMs.
- ☐ Further, in order to compensate the losses on account of flexible operation, CERC has already incorporated most of the recommendations of CEA's compensation methodology such as for additional Capex, oil consumption, additional auxiliary power consumption, and heat rate degradation.
- ☐ Therefore, all generating utilities are requested to complete the retrofit, control system tuning, trial runs etc. of all units under Phase I by June 2026 in consultation with the OEM positively.
- ☐ It is also requested to furnish the progress and updates (as per attached format) of First Phase units by the end of November 2025 and thereafter every month.

As per 56th TCC meeting

DVC informed that 40% MTL can be achieved only with higher grade of coal. As coal is arranged from various sources having different grades, the sustained operation of boilers at 40% MTL with different grades of coal has not yet been demonstrated.

DVC representative further highlighted that many of the state generators has not yet implemented 55% technical minimum norm.

56th ERPC Decision:

- ✓ ERPC advised all thermal generating stations to intimate their preparedness as well as technical difficulties, if any, in achieving 40% MTL to ERPC Secretariat. ERPC Secretariat would subsequently forward the issues to CEA.
- ✓ The forum also advised to ER states to operate intra-state TPPs at MTL(55%) to assist in grid stability in view of increasing RE penetration and to introduce compensation mechanism for operation of intra-state thermal generators at technical minimum through respective SERCs.

As per 241st OCC:

WB SLDC stated that compensation mechanism for operating at 55% technical minimum has been incorporated in the recently drafted tariff regulations of WBERC and accordingly necessary changes shall be done in the SAMAST scheduling software for enabling all intra-state generators (WB) to operate at 55% technical minimum as per respective schedules.

WBPDCCL confirmed technical readiness to operate at the compensated level; the earlier outstanding issue of compensation for operation below normal technical minimum has since been addressed in the new regulation.

241st OCC Decision

- ✓ OCC forum was apprised that most inter-state generating stations are presently operating at around 55% technical minimum, and are now required to move towards 40% MTL flexible operation. It is also advised to state generating stations to operate at 55% Technical minimum level.
- ✓ All thermal generators were advised to furnish specific operational difficulties, if any, in operating at 40% technical minimum to ERPC secretariat before the ensuing NPC meeting (scheduled in September 2026).

Deliberations in the meeting

***DVC** submitted the present status of Flexible Operation at BTPS Unit #A (500 MW), RTPS Unit #2 (600 MW) & MTPS Unit #7 (500 MW) as well as presented the technical limitations of running at 40% MTL including poor coal quality, Boiler metal temperature excursions during ramp-up and ramp-down operations, Flame stability issues and Continuous operation at 40% MTL resulting in increased thermal & mechanical stress on boiler and turbine components. (details enclosed at **Annex B.2.1.(b)**)*

West Bengal SLDC informed:

- *West Bengal Electricity Regulatory Commission (WBERC) has recently notified regulations providing for compensation to generating stations operating down to 55% technical minimum. However, the regulation leaves the detailed modalities (compensation formula, ramp rate, and related commercial parameters) to be worked out by the State Load Dispatch Centre (SLDC).*

- WBSLDC has been given four months by WBERC to prepare this modality, after which it will be submitted to State regulatory Commission for approval; only after such approval can 55% technical-minimum operation with compensation commence in the state.

OCC Decision

- OCC welcomed West Bengal's initiative and suggested other constituent states to move towards similar regulatory provisions, noting this would support generators financially while enabling greater flexible operation as renewable energy (RE) penetration increases in the grid.
- WBSLDC was advised to finalize the detailed modalities for compensation and get it approved by WBERC within the stipulated four-month period. This compensation-mechanism may be finalized in line with prevailing ISGS practice.
- All thermal generators were again advised to furnish specific operational difficulties, if any, in operating at 40% technical minimum to ERPC secretariat before the ensuing NPC meeting (scheduled on 18th September 2026).

c) Upgradation of 66/11 kV Bulbuley Sub-station to 132 kV Level and Associated Transmission Infrastructure works in Gangtok

Upgradation of 66/11 kV sub-station at Bulbuley (Gangtok) to 132 kV voltage level, establishment of AIS extension bay at POWERGRID 132/66 kV sub-station at Lagyap and GIS bays at Bulbuley along with installation of 132/66 kV, 2x25 MVA transformers at Bulbuley along with construction of 132 KV LILLO arrangement at 132 kV D/C transmission line. Estimated Cost = Rs. 167.42 Crore. Bulbuley is the only major substation within Gangtok proper that still has scope for upgradation and availability of ROW corridor for construction of transmission line. Capacity augmentation is not possible in remaining two other 66/11 kV sub-stations within Gangtok proper, i.e. Tadong and Sichey, owing to space and ROW constraints. These areas are now completely built up and commercialized. Urgent need to upgrade the main power supply to main Gangtok city area to 132 kV voltage level due to current condition of the existing lines all of which are 66 kV lines. Need to undertake the work soon since there will not be any ROW availability or space for expansion of sub-station in Gangtok proper in the near future.

Deliberation in 56th TCC meeting:

MS, ERPC informed that as per CEA recommendation, instead of upgrading Bulbuley S/s to 132 kV level, Perbing S/s has been recommended for early commissioning to cater to the load of Gangtok area.

Powergrid informed that the Perbing S/s will be commissioned by **5th June, 2026**.

56th TCC Decision:

TCC noted.

56th ERPC Decision:

ERPC noted.

As per 241st OCC:

PowerGrid updated:

The Perbing S/S has already been charged at 132 kV level. ICT has also been charged. Charging at 66 kV level is still pending that shall be done by **31st July 2026**.

241st OCC Decision
OCC noted.

Deliberations in the meeting

Powergrid apprised:

The ICT at Perbing S/S is charged but not loaded at present as 11 kV distribution feeders have not been charged by Sikkim.

OCC Decision:

OCC agreed to convene a special meeting involving ERPC, PowerGrid and Sikkim.

d) Restoration of 220KV FSTPP LALMATIA Line.

The line is out of service since long due to tower collapse. Presently 220 kV Farakka-Lalmatia line is charged(from loc no 241 to loc 84) at 132 kV voltage level for anti-theft purpose by tapping at loc. No. 100-101.

As per 56th TCC :

- ✓ JUSNL informed that the line will be completed by 10th June.
- ✓ ERLDC suggested that JUSNL may take all requisite FTC clearances beforehand so that the line may be charged at the earliest.

56th TCC Decision:

TCC noted the work progress by JUSNL and advised NTPC to check healthiness of the bay equipment of the line at Farakka end.

56th ERPC Decision:

ERPC noted.

As per 241st OCC:

NTPC submitted:

- Through Agreement dtd. 25.10.2021(attached at **Annex B.2.1.f**) Farakka–Lalmatia Transmission System (FLTS) was handed over to JUSNL by ECL on “as is where basis at ZERO cost and relieving NTPC from O&M responsibility.
- As per the agreement, JUSNL shall have the ownership of FLTS and shall maintain FLTS at their own cost. Responsibility for operation and maintenance shifts to JUSNL.
- JUSNL will bear all costs related to Overhauling, Upgradation ,Replacement ,Maintenance and any incidental expenses.

Communications were sent to JUSNL as per details below:

- Vide letter dated 31.01.2026, NTPC requested JUSNL to update the bay equipment and protection system along with the new transmission line. Requisite AC and DC supply source for the Bay and CRP will be made ready by NTPC.
- Vide letter dated 14.02.2026, NTPC requested JUSNL to update the complete bay equipment

and protection system along with the new transmission line. Necessary AC and DC supply source for the Bay and CRP will be made ready by NTPC.

241st OCC Decision

- ✓ The forum advised JUSNL and NTPC to decide the ownership of bay at Farakka end bilaterally. Accordingly, bay works at Farakka end may be initiated at the earliest.

Deliberations in the meeting

JUSNL updated:

- ✓ *Internal administrative approval for restoring the bay at FSTPS end has been obtained and necessary work has been taken up, involving replacement of CT/PT, circuit breaker and associated equipment.*
- ✓ *The 220KV FSTPP LALMATIA Line is presently anti-theft charged.*
- ✓ *Management has agreed to take ownership of the bay.*
- ✓ *Spare Godda bay equipment material(at 220 kV Godda S/S)will be used for the purpose of bay upgradation at NTPC Farakka end and the restoration work is expected to be completed within 2–3 months.*

OCC Decision

OCC noted the progress and advised JUSNL to expedite completion of bay upgradation at Farakka end within 3 months in view of criticality of the 220KV FSTPP LALMATIA Line for Farakka islanding scheme. NTPC agreed to extend all necessary assistance to JUSNL in this regard.

e) A review of the Flue Gas Desulfurization (FGD) installation, in consultation with beneficiaries, is required for ongoing and operational Category 'C' thermal power projects of NTPC: GRIDCO

- ☐ It is to mention that MoEF&CC has notified Environment (protection) Rules on 11.07.2025 wherein applicability of SO₂ emission standards in thermal power plants has been revised based on their location/category. As per the above amendment, Sulphur dioxide emission standards shall not be applicable to all Category C thermal power plants subject to ensuring compliance of stack height criteria notified vide notification number GSR 742 (E) dated 30th August 1990.
- ☐ As understood CEA vide letter dtd 20.08.2025 intimated all RPCs that the generating utilities would be required to review the FGD installation in consultation with Discoms at the ongoing projects at Category C TPPs.
- ☐ In the said letter, CEA referred a meeting that was held with GENCOs on 18.07.2025 and with vendors on 08.08.2025 with a request to RPCs to convene a special TCC/RPC meeting at the earliest and deliberate the matter and various proposals furnished by CEA.
- ☐ Further, as per the minutes of the meeting Dtd 25.07.2025, the generating companies were advised to discuss various aspects with the beneficiary Discoms their willingness to have FGD installed and pay for the cost attributable to FGD (CAPEX+OP-EX) or stop the project and pay the fixed cost (CAPEX) incurred so far.
- ☐ In this regard this is to be mentioned that, Darlipalli STPS of NTPC is coming under category 'C' as per MOEF Notification dtd 11.07.2025. FGD has been already installed and commissioned from 01.07.2024 for unit#2 and 27.01.2025 for unit#1.

NTPC has been raising bills approx. 1.5 Crs/month to 1.8 Crs/month towards secondary energy

charges (OP-EX) as Odisha has 50% allocation from the said project. Hence, GRIDCO vide letter dtd 02.09.2025 requested NTPC for discontinuation of supplementary energy charge (ECS) towards FGD for Darlipalli STPS to which NTPC replied that “FGD system has been declared Commercial Operation in accordance with the provision of CERC Tariff Regulation 2019, and the bills of supplementary variable charges are also being raised accordingly to all the constituent beneficiaries of the station. The matter is being looked into by concerned statutory authorities and any decision and subsequent direction issued in such cases will be duly complied by NTPC.”

In view of the exemption for Category 'C' TPPs from SO₂, standards GRIDCO raises the following points for discussion:

1. Discontinuation of FGD Operation: NTPC is requested to stop the operation of the FGD system at Darlipalli STPS, as it is no longer mandatory under the revised MoEF&CC notification.
2. Assessment and Financial Justification: NTPC should furnish the project-wise status of FGD implementation for all its generating stations in the Eastern Region. Furthermore, a detailed financial assessment must be conducted to compare the cost benefit of operating with FGD versus without FGD; ensuring beneficiaries are not burdened with unnecessary CAPEX and OP-EX.

Deliberation in 56th TCC meeting

- ✓ NTPC stated that the units where FGD has already been commissioned, not keeping the FGD operational might result in deterioration of the equipment.
- ✓ WBSEDCL suggested that NTPC may run FGD on intermittent basis to prevent such an outcome and not burden the beneficiaries with higher OPEX that results due to continuous operation. The matter of aux consumption issue will be resolved by adopting suitable methodology.

56th TCC Decision:

TCC advised NTPC to adhere to the decision taken in the meeting held on 12.12.2025, chaired by Secretary, Power GoI, for TPPs which have already installed FGD, where it has directed that the future course may be mutually decided by respective stakeholders. TCC opined that the matter, being a national level issue, may be referred to NPC and referred the matter to ERPC for concurrence.

Deliberation in 56th ERPC meeting

- ✓ GRIDCO emphasized that NTPC needs to discontinue the operation of the FGD system at Darlipalli STPS immediately, in strict compliance with the MoEF&CC Environment (Protection) Rules dated 11.07.2025. Such unilateral operation of FGD by NTPC Darlipalli is imposing an unjustified financial burden on GRIDCO and subsequently on consumers.
- ✓ WBSEDCL agreed with the submission of GRIDCO and stated that NTPC should discontinue the operation of the FGD system at Darlipalli STPS. As an alternative measure, WBSEDCL suggested that FGD can be run intermittently to keep the FGD in serviceable condition. They also submitted that without any clear directives to run the FGD, the same may attract audit observations.
- ✓ NTPC representative informed that keeping the FGD system idle for a prolonged period (6 months) might result in deterioration of the FGD components.
- ✓ Member (GO&D), CEA opined that as CAPEX is already being paid by the constituents and since OPEX constitutes a relatively small portion of overall tariff, NTPC may be allowed to run the FGD as it helps in reduction of the pollution level in the nearby area of power plant and also

keep the FGD in working condition.

ERPC Decision:

ERPC forum suggested that NTPC should submit the cost-benefit analysis report of running the FGD by considering the sale of gypsum to ERPC Secretariat by 30th June, 2026 and the matter will be deliberated in sub-committee meeting of ERPC for taking the further course of action.

As per 241st OCC:

NTPC informed:

Report on Cost-benefit analysis for FGD installation is under approval stage at Corporate Commercial and the same shall be shared in the next OCC.

241st OCC Decision

NTPC was advised to positively submit cost benefit analysis along with inputs from concerned OEM at the earliest.

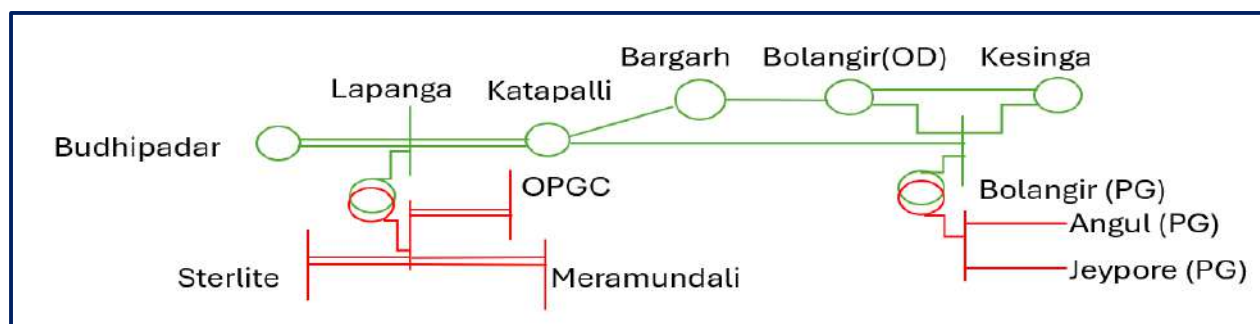
Deliberations in the meeting

OCC Decision

NTPC was advised to expedite submission of the pending cost-benefit analysis with OEM inputs regarding continued operation/FGD compliance of Category-C thermal units.

f) High loading and N-1 violation of 400/220kV ICTs at Bolangir (POWERGRID):

Bolangir S/S is one of the major 400/200kV S/S in western Odisha catering load to Bolangir, Kesinga, Bargarh & Katapalli areas. In recent part, demand in this part of Odisha has increased significantly, which ultimately impacts the loading of ICTs of Bolangir. Supporting source of these areas is 220kV-Lapanga-Katapalli -D/C, which also operating at high loading mainly during summer period. ICT loading further increases with opening of 220 kV Katapalli-Bolangir(PG) and 220kV Katapalli-Bargarh lines to limit the loading of 220 kV Budhipadar-Lapanga D/c and 220 kV Lapanga-Katapalli D/c which are not N-1 compliant.



As per 56th ERPC meeting

Chief Engineer, CEA informed that upgradation of this line has already been approved under RA plan for 2026-27. Further, he also pointed out that in transmission RA plan of Odisha, the reconductoring

of intra-state lines is to be completed by OPTCL in the timeframe of 2026-27. However, there is no progress in this regard. CTUIL also submitted the delay in strengthening of intra-state networks by OPTCL will result in ISTS system vulnerable in future.

56th ERPC Decision:

The forum advised OPTCL to plan the reconductoring of lines as proposed in the RA plan for 2026-27 at the earliest and submit a timeline of the same to ERPC secretariat.

As per 241st OCC:

OPTCL informed that proposal of reconductoring of 220 kV intra-state lines connecting Bolangir(PG) is still in planning stage.

241st OCC Decision

- ✓ OCC opined that persistent delay in strengthening of intra-state network of Odisha may render the ISTS network vulnerable in near future and thereby took a serious note on the issue.
- ✓ OCC urged OPTCL to submit a firm timeline and detailed work schedule for reconductoring of associated 220 kV lines from Bolangir(PG) S/S in the next OCC.

Deliberations in the meeting

OPTCL informed

- The root cause of recurring overloading/manual tripping on certain lines (e.g., in the Bolangir area) was attributed to loading imbalance pending completion of a parallel system-strengthening (S2S) line, expected to relieve the situation once commissioned.
- *Proposal for an additional double-circuit line between Bolangir(PG) and Bolangir (OPTCL sub-station) is recently placed and is expected to be progressed soon.*
- Along with 220 kV lines, reconductoring of 132 kV lines from Bargarh & Katapalli is also taken up.
- Price bids for reconductoring of the lines: Budhipadar-Lapanga and Lapanga-katapalli has been opened on 10th August 2026 and order placement is expected within next 30 days after bid evaluation. Reconductoring of 400 kV Meramandali-mendhasal line is also taken up in this project.
- HTLS reconductoring of all relevant lines is expected to be completed within next 10 months.

Detailed status of HTLS reconductoring enclosed at **Annex B.2.1.f** 

OCC Decision:

OCC forum urged OPTCL to place orders for reconductoring works promptly on evaluation of price bids and to furnish periodic progress reports to ERPC. OPTCL has to ensure HTLS upgradation of all the associated transmission lines before the onset of next Summer. This is essential for both ISTS network and intra-state network of Odisha.

g) Diversion of RPC approved Spare Transformers and Reactors to the constituents / State Transmission Utilities

- ☐ **Chairperson, CEA/NPC** opined that private Transmission Service Providers (TSPs) may be considered for diversion of RPC approved spare transformers and reactors on a case-to-case basis, subject to discussion in the respective Regional Power Committee (RPC) forum.
- ☐ **Chairperson, CEA/NPC** further suggested that a suitable mechanism for periodic rotation/usage of spare transformers and reactors within the utility or among the constituents / State Transmission Utilities (STUs) may be explored. This would ensure that spare transformers and reactors do not remain unused for prolonged periods and become too old without operational utilization.

- It was suggested that all RPCs may review the existing Standard Operating Procedures (SOPs) prepared by SRPC and NRPC in their respective forums, obtain comments from stakeholders, and submit their suggestions. The matter may thereafter be deliberated in the Operation Sub-Group of the NPC for finalization of a uniform SOP at the national level.

As per 17th NPC decision:

All RPCs to explore a suitable mechanism for periodic rotation/usage of spare transformers and reactors within the utility or among the constituents / State Transmission Utilities (STUs) and also examine the inclusion of Private Transmission Service Providers in the scheme. All RPCs may also review the existing Standard Operating Procedures (SOPs) prepared by SRPC and NRPC in their respective RPC forums.

56th TCC Decision:

Forum suggested all the utilities to submit their comments regarding the diversion of spares as well as the penal provision to ERPC secretariat at the earliest so that the matter may be deliberated in the next NPC.

As per 56th ERPC meeting

WBSLDC suggested that the penalty may be levied after duration of 36 months as procurement of new ICTs is taking around 3 years.

Further, representative of Powergrid suggested that penalty of 15% of approved tariff may be adjusted in the procurement of new spare ICTs & reactors.

Member Secretary stated that the spare available in the regional pool should be used in case of exigency and the state should maintain their spare as per extant norms of CEA/CERC.

56th ERPC Decision:

After detailed deliberation, Forum decided that all the concerned utilities will submit their inputs on draft SOPs to ERPC secretariat by 30th June, 2026. The input of ERPC based on the feedbacks received from concerned stakeholders will be submitted in the forthcoming NPC.

As per 241st OCC:

Regarding the diversion of spares including penal provisions, WBSETCL submitted comments as stated below:

- 1) Date of effect of penalty related order, if any, should be after the issuance of such order only. It should not be effective retrospectively.
- 2) It would be prudent that the penalty may be levied after a duration of 36 months as procurement of new ICTs is taking around 3 years.
- 3) A penalty of 10% of approved tariff for the diverted equipment may be imposed on the borrower for the duration of deployment beyond 36 months.
- 4) Periodic energization/testing of idle spares may also be considered to prevent deterioration.
- 5) Collected penalty amounts may accrue in any separate reliability fund for proper use in future for necessary inspection, testing etc.
- 6) After release of equipment by the constituents, the equipment may be kept as spare in the nearest location, if possible. It will reduce the cost of transportation otherwise.

241st OCC Decision

- ✓ All ER utilities (except WBSETCL) were advised to share their inputs/comments (if any) on the SOP on diversion of spares to ERPC by **31st July 2026**.
- ✓ The forum opined that the respective inputs will be placed in the ensuing NPC meeting for consideration and subsequent finalization of the SOP.

Deliberations in the meeting

*The draft SOP prepared by NPC for diversion of RPC approved Spare Transformers and Reactors to the constituents was presented to the forum with suitable modifications (enclosed at **Annex B.2.1.g**).*

OCC Decision

OCC agreed in principle with a maximum diversion period of 30 months from the zero date of diversion, extendable by a further 12 months in case of exigency with RPC approval and technical justification (i.e., 30+12 months formulation), superseding the earlier 24-month proposal. The guideline, including MOU execution timelines, inspection requirements, and penalty provisions, is to apply prospectively and it will be formally adopted after approval in the ensuing NPC meeting.

h) Update on islanding schemes in ER.

The list of existing Islanding schemes in ER is as under:

Eastern Region (ER)		
1	Kolkata (CESC) IS	City/Major Town/ Strategic Load
2	Howrah (Bandel) IS	City/Major Town/ Strategic Load
3	Bakreswar TPS IS	Industrial and Railway load
4	Haldia (Tata Power) IS	Industrial areas of Haldia and Port
5	Chandrapura (CTPS DVC) IS	Industrial areas

As per 238th OCC:

OCC suggested for formation of a sub group committee under chairmanship of S.E. (Operation), ERPC comprising members from ERPC, ERLDC, concern SLDC and Generating company for regular monitoring of islanding scheme formation and implementation in the ER States. **The committee shall coordinate with stakeholders and submit progress report in OCC.**

A special sub-group has been constituted for the identification and implementation of islanding schemes in the capital cities of ER states.

56th ERPC Decision:

The forum advised that the matter may be regularly followed up in OCC meetings.

1. Ranchi Islanding Scheme

- ☐ Nomination for ER sub-group of islanding scheme implementation was received from JUSNL.
- ☐ The kick-off meeting for the implementation of the Ranchi City Islanding Scheme was held on **23rd April 2026** under chairmanship of ERPC.

- To accurately define the electrical boundaries and load behaviour of the island, SLDC Jharkhand was requested to share:
- **Island Boundary & Feeder Logic:** Clearly define the islanded area and provide a comprehensive list of feeders that must be disconnected to ensure successful island formation.
- **Load Profiles:** Node-wise load data (Maximum, Minimum, and Average) for the identified area to ensure generation-load balancing.
- **Defense Mechanism Data:** Current node-wise loads of Jharkhand that are already covered under **AUFLS** (Automatic Under Frequency Load Shedding) and **ADMS** (Automatic Demand Management System).
- **Infrastructure Timeline:** Confirmation of the expected commissioning date for the **400kV Latehar New – Patratu New D/C** line, as this must be integrated into the islanding model.

As per 56th TCC meeting

JUSNL informed that they have already provided data regarding Ranchi islanding scheme.

As per 241st OCC:

ERLDC highlighted some constraints faced in the dynamic study of Ranchi islanding scheme, such as, low voltage issues being faced due to feeding large load segment. This is due to inadequacy of power evacuation corridor from PVUNL. Presently, only few 220 kV lines are feeding all the identified loads and thus, getting overloaded. 400 kV Patratu-Latehar line needs to be commissioned for N-1 compliance in islanding condition.

241st OCC Decision

- ✓ The forum advised ERLDC to carry out dynamic study considering the upcoming 400 kV Patratu-Latehar line and recommend the study results within a week.
- ✓ JUSNL was advised to expedite commissioning of 400 kV Patratu-Latehar line and submit detailed execution plan in next OCC.

Deliberations in the meeting

ERLDC submitted

- ✓ *A detailed dynamic feasibility study has been completed jointly with the State Load Despatch Centre (SLDC), Jharkhand and the report has been shared with SLDC. Patratu generating station has been considered in the islanding scheme.*
- ✓ *A load-shedding/stabilization scheme was also proposed as part of the study to keep the island stable under contingency conditions*
- ✓ *The next step is preparation of the Detailed Project Report (DPR) by the state, followed by ERPC approval and PSDF Committee approval and, before implementation can proceed.*
- ✓ *The 400 kV Patratu-Latehar line was highlighted as critical to island stability, since the Ranchi island currently relies only on 220 kV lines, which get overloaded; the 400 kV line will relieve loading and improve voltage stability, and thus, needs to be commissioned before the islanding scheme itself. It was emphasized that commissioning of 400 kV Latehar–Patratu (New) D/C is a pre-requisite for implementation of the islanding scheme, considering its importance for voltage stability of the island.*

JUSNL informed that construction of the line has been awarded to PGCIL as a deposit work.

Powergrid ER-I updated that ownership of the line has been settled and completion of the 400 kV Patratu-Latehar line is expected within a year, tentatively by March 2027.

OCC Decision

- *SLDC Jharkhand was advised to finalize the DPR of Ranchi islanding scheme in consultation with ERLDC and place it for approval of ERPC so that sanction of funds from PSDF can be obtained at the earliest. Commissioning of 400 kV Latehar–Patratu (New) D/C shall be appropriately considered while finalizing the implementation sequence of the Ranchi Islanding Scheme.*
- *Powergrid ER-I was urged for expedited commissioning of the 400 kV Patratu-Latehar line, given its criticality for the Ranchi islanding scheme.*

2.Patna Islanding Scheme under PSDF

- *In 54th TCC meeting held on 23.06.2025, BSPTCL had proposed to implement Patna Islanding Scheme through Internal Resource Fund.*
- *However, a meeting was held on 24th June 2025 under the chairmanship of the Hon'ble Minister of Power and Housing & Urban Affairs, wherein the matter of Islanding Patna city was discussed. In the meeting, it was decided that the State of Bihar would submit a proposal for funding the Islanding scheme by the Ministry of Power).*
- *In compliance to minutes of the meeting held on dt. 24.06.2025, Board of Directors, BSPTCL has approved for implementation of Patna Islanding Scheme through PSDF in 131st BOD meeting held on dt. 17.07.2025 vide its resolution no. 131-06.*
- *In line with the above, a proposal has been submitted for Implementation of Patna Islanding Scheme under PSDF to NLDC vide this office letter no. 549 dt. 18.07.2025 along with all the required documents in compliance to minutes of meeting held on dt. 24.06.2025.*
- *Further, Chief Engineer (Generation), CEA has requested the recommendation of ERPC for implementation of Patna Islanding Scheme through PSDF.*

As per 56th TCC meeting

SLDC Bihar informed that the PSDF Appraisal committee approved the PSDF grant proposal of Patna islanding scheme and the approval from the PSDF monitoring committee is awaited.

As per 241st OCC:

SLDC Bihar updated that in-principle approval has been obtained from NLDC on 10th June 2026 while sanction of funds is still awaited.

241st OCC Decision

OCC noted.

Deliberations in the meeting

SLDC Bihar informed :

- ✓ *In-principle funding approval has been received from PSDF. Also, it has been clarified by NLDC on 28th July that the Letter of Award (LOA) can be issued without waiting for final PSDF sanction.*
- ✓ *Accordingly, internal approvals have been obtained and the LOA is to be placed by 25th August 2026.*
- ✓ *Implementation is estimated to take about 15 months thereafter.*

OCC Decision

OCC noted the progress. Bihar was advised to adhere to the timeline of placing the LOA (by 25th August'26) as well as ensure timely implementation of Patna Islanding Scheme.

3. IB Valley TPS Islanding Scheme (Odisha)

IB valley TPS Islanding scheme has also been put on hold for long time. The status regarding the same has been sought on urgent basis by Ministry of Power (Govt of India).

As per 56th TCC meeting

OPGC informed that those necessary lines have been identified for Ib valley islanding scheme but base data is required from SLDC Odisha for further action.

As per 241st OCC:

SLDC Odisha informed that planning of Bhubaneswar Islanding Scheme (meant for Bhubaneswar & Cuttack area) is being taken on priority as of now. Once the Bhubaneswar islanding scheme gets implemented, IB valley TPS islanding scheme will be taken up for further proceedings.

241st OCC Decision

OCC noted.

Deliberations in the meeting

It was apprised that OPGC will take up this islanding proposal for study and implementation, the same is not under priority now.

OCC Decision

OCC noted.

4.Bhubaneswar Islanding Scheme (Odisha)

As per 241st OCC:

SLDC Odisha updated:

- Work order for preparation of the Detailed Project Report (DPR) has been issued and a kick-off meeting is planned within the next two weeks.
- Three generating units have tentatively been identified as potential islanding sources — two units of GMR and one unit at NTPC Kaniha — though final selection is pending.
- The DPR preparation is expected to take about 3–4 months once the generation source is finalized.

241st OCC Decision

SLDC Odisha was advised to finalise the generation source for the Bhubaneswar–Cuttack–Puri scheme and share the exact load-shedding quantum with ERPC/ERLDC before proceeding to DPR finalization.

Deliberations in the meeting

OPTCL updated:

- ✓ *The earlier proposal to island using a small 65 MW generator was assessed as inadequate.*

- ✓ A larger generating station in the vicinity— GMR Kamalanga (2×350 MW) was identified .The sole STU-connected unit of GMR KEL has been proposed as the generating station, with around 300 MW load of Bhubaneswar, Cuttack and Puri proposed for inclusion in the island.
- ✓ PRDC has been assigned to carry out the detailed studies (load flow, short-circuit, dynamic stability, frequency response) and prepare the DPR.
- ✓ Meeting with DISCOMs is under progress for identification of essential loads and associated schematic single-line diagrams (SLDs).
- ✓ Sambalpur islanding is also proposed (with 2×24 MW units of Chipilima + units of Burla, if available) hydro station as support.

ERLDC requested SLDC ODISHA to share the study results with ERPC & ERLDC considering GMR-STU unit of 350MW as the source supplying the island, for further validation.

OCC Decision

OPTCL was advised to complete the technical studies (load flow, short-circuit, dynamic stability, frequency response) for the Bhubaneswar and Sambalpur schemes and share the study results with ERPC & ERLDC .

5. Farakka islanding scheme

This is presently not in service due to long outage of 220 kV Farakka-Lalmatia line.

241st OCC Decision

Implementation of Farakka islanding scheme shall be taken up after restoration of 220KV FSTPP LALMATIA Line.

Deliberations in the meeting

OCC Decision

Implementation of Farakka islanding scheme shall be taken up after on-load charging of 220KV FSTPP LALMATIA Line. JUSNL was advised to expedite the same.

i) Review of AUFLS in Eastern Region: SCADA Integration & Data Updation

- ◆ Based on the recommendation and decisions in 14th NPC meeting held on 05.02.24, 214th OCC meeting and special meeting on 10.07.2024, a load relief quantum of 6916 MW was finalized for Eastern Region. UFR Feeders real time monitoring has been discussed in NPC as well as various fora of ERPC.
- ◆ Further, with new IEGC 2023 the same has been mandated as quoted below: IEGC 2023, Clause 13.d: “SLDC shall ensure that telemetered data of feeders (MW power flow in real time and circuit breaker status) on which UFR and df/dt relays are installed is available at its control centre. SLDC shall monitor the combined load in MW of these feeders at all times.
- ◆ SLDC shall share the above data with the respective RLDC in real time and submit a monthly exception report to the respective RPC. RLDC shall inform SLDCs as well as the concerned RPC on a quarterly basis, durations during the quarter when the combined load in MW of these feeders was below the level considered while designing the UFR scheme by the RPC. SLDC shall take

corrective measures within a reasonable period and inform the respective RLDC and RPC, failing which suitable action may be initiated by the respective RPC.”

As per 56th TCC meeting

Both Bihar & Jharkhand updated that they will Complete AUFLS implementation as well as SCADA integration by end of June 2026.

56th TCC Decision:

TCC urged early completion of AUFLS implementation as well as SCADA integration.

56th ERPC Decision:

ERPC advised concerned ER states for early completion of AUFLS as well as SCADA integration.

Update:

The load relief quantum to be implemented at each stage of AUFLS for FY 2026-27 is computed by NPC based on the regional peak demand met in FY 2025-26.

$$\text{Stage-wise Load relief} = \text{Stage-wise relief (\%)} \times \text{RPRegion} \times \text{AP/RP}$$

AP: All India Peak Demand Met in MW

RP: Regional Peak Demand Met

As per this methodology, total load relief quantum for ER comes out as 7195 MW against 6916 MW in preceding year.

This 7195 MW of load relief has to be distributed among all ER states as per methodology to be finalized in OCC forum.

As per 241st OCC meeting:

SLDC Bihar updated that identification of new feeders for the revised load relief quantum is under approval stage

241st OCC Decision

- ✓ All SLDCs were advised to ensure implementation of adequate stage-wise load relief quantum as detailed below by August 2026. Entire number of feeders should also be integrated in SCADA.

Latest status as follows:

Control Area	Required Load Relief					Implemented Load Relief				
	Stage –I (49.4 Hz)	Stage –II (49.2 Hz)	Stage– III (49.0Hz)	Stage– IV (48.8Hz)	Total Relief by Control Area	Stage – I (49.4 Hz)	Stage –II (49.2 Hz)	Stage– III (49.0Hz)	Stage– IV (48.8Hz)	Total Relief by Control Area
	(MW)	(MW)	(MW)	(MW)		(MW)	(MW)	(MW)	(MW)	
BSEB	345	399	466	466	1676	316	373	409	424	1522
JSEB	103	130	151	151	535	90	103	157	129	479
DVC	159	208	243	243	853	171	214	203	239	827
Odisha	304	372	434	434	1544	374.5	377.5	370	452	1574
WB & CESC	523	613	715	715	2567	560	574	712	715	2561
Sikkim	5	5	5	5	20	0	0	0	0	0

TOTAL	1439	1727	2015	2015	7194	1508.5	1642.5	1848	1927	6963
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Deliberations in the meeting

- ✓ SLDC Bihar apprised that additional load quantum is communicated to DISCOMs but receipt of feeder list is awaited. Once feeder list is received, revised load relief quantum will be implemented within 1 week.
- ✓ SLDC Jharkhand intimated around 1 month time for completion.
- ✓ DVC and Odisha informed that the implemented load relief quantum exceeds the requirement in almost all stages.
- ✓ Citing an approximate 3.6% load-growth, WB SLDC informed implemented quantum is already more than the requirement.

OCC Decision

- ✓ All SLDCs were advised to ensure implementation of adequate stage-wise load relief quantum as detailed below by 10th September 2026. Entire number of feeders should also be integrated in SCADA within the stipulated timeline.

2.2 Issues for follow-up: ERLDC

a) Replacement of CLR insulators of PMJTL lines

- Replacement of faulty Decan make CLR insulators in the 765 kV New Ranchi–Medinipur–New Jeerat corridor and associated 400 kV PMJTL lines has been under discussion in various OCC/TCC meetings owing to the strategic importance of these transmission corridors for reliable power supply to Kolkata and adjoining areas of South Bengal. While replacement activities in the 400 kV system have substantially reduced outage incidences, frequent shutdown requests continue to be received for the 765 kV corridors.
- During FY 2025-26, POWERGRID availed 63 shutdowns on the 765 kV New Ranchi–Medinipur–New Jeerat corridor for replacement of failed CLR insulators. Further, during the ongoing summer period, several additional emergency shutdown requests have already been processed. Such frequent outages, particularly during periods of high demand in West Bengal, pose operational challenges and increase the risk to reliability and security of power supply to Kolkata and surrounding areas. Shutdown of one circuit followed by any contingency on the parallel circuit may lead to a highly stressed system condition.
- In 239th POWERGRID informed that around 5,000 faulty Decan make CLR insulators in the 400 kV system have been replaced, significantly reducing trippings compared to the previous year. However, procurement of replacement insulators for the identified 765 kV corridors has been delayed, with completion now targeted by October 2026. OCC directed POWERGRID to expedite replacement of defective insulators, undertake comprehensive condition assessment of vulnerable locations, improve shutdown planning through consolidated weekly outages, and minimize emergency shutdowns, particularly during the summer.

As per 240th OCC meeting:

Powergrid informed:

- Approximately 28,000 insulators (400 & 765 kV combined) are planned for replacement. For 400 kV system: about 10,000 insulators to be replaced and for 765 kV system: about 17,000 insulators to be replaced.

- Total 10273 Nos (400KV) and 17785 Nos (765KV) CLR insulators are under procurement. As of 30th June 2026, 700 (at 400 kV level) and 850 (at 765 kV level) insulators have been supplied at site and progressive supply will continue.
- Replacement work will be prioritized on critical lines (e.g. 765 kV D/C Ranchi- Medinipur) and shall be carried out progressively, aiming to complete both supply and replacement by October-November 2026.

OCC opined that a special meeting may be convened by ERPC involving stakeholders from Powergrid, WB & ERLDC to finalize a detailed plan for replacement of CLR insulators in a phased manner.

In **241st OCC**, Powergrid updated the detailed Plan as well as status for replacement of defective CLR insulator in PMJTL elements. As on 08th Jul'26, 2680 nos. of 400kV insulators and 1295 nos. of 765kV insulators were replaced as reported by POWERGRID. The Insulator replacements in 400kV and 765kV were scheduled to be completed by Oct'26 and Dec'26 as per tentative timelines submitted by POWERGRID.

Deliberations in the meeting

Powergrid submitted :

- ✓ *Industry-wide constraints in supply of 400 kV & 765 kV insulators were flagged. Due to limited availability of vendors, and corresponding limitations in the supply of 765/400 kV insulators, they are presently unable to advance deliveries beyond the existing LOA schedule. TS of the insulators has been suitably modified, considering CLR material and design of tower also changed accordingly.*
- ✓ *Present progress of work is slow amid inclement weather conditions. Following the ICT outage at New Jeerat, replacement of the existing insulators with Non Deccan make insulators at Medinipur & New Jeerat S/S has been taken up on priority basis for which, insulators are being diverted from Satna S/S for replacement works at both ends.*
- ✓ *After the lot of new insulators of sufficient quantity is received in September'26, work will resume in 400kV New Jeerat-Subhasgram Ckt-1 on priority.*
- ✓ *Detailed plan for procurement and replacement of insulators attached at **Annex B.2.2.a***
- ✓ *A request was also made that shutdowns which are purely internal in nature and do not affect any other constituent's activity or grid-security constraints may be granted faster approval — at least one week before the activity. The shut-down of lines for Insulator Replacement work that are being submitted 1(one) **week in advance may be treated under OCC approved line shut-down category.***

OCC Decision

Powergrid shall continue replacement of the defective insulators on priority and furnish weekly progress/status updates to ERPC/ ERLDC. The final schedule of shutdown and replacement shall be shared immediately after availability of the required insulators.

b) Restoration of 400kV Patna-Naubatpur-Balia Ckt#1 to its original configuration

- Bypassing of 400kV Patna-Naubatpur Ckt#1 and 400kV Naubatpur-Balia Ckt#1 at Naubatpur(BGCL) was accorded in 206th OCC meeting held on 31.08.2023 for drawl of start-up power of 2x660 MW Buxar Thermal Power Project by connecting one circuit of the Buxar–Naubatpur line using the 400kV Balia-1 bay at Naubatpur. Subsequently, with the interim arrangement, 400KV

Buxar–Naubatpur Ckt#2 was charged on 11.05.24 and Buxar started drawing start up power. After commissioning of Buxar Unit#1, the interim arrangement was continued.

- ❑ The termination bays (2 nos) of Buxar at Naubatpur were ready in April'26. Thereafter, two nos. of 400kV lines were terminated at Naubatpur in months of May & June 2026. However, restoration of 400kV Patna-Naubatpur-Balia Ckt#1 bypassed at Naubatpur to its original LILO configuration was not followed upon. As Buxar connectivity at 400kV level with Naubatpur is commissioned now, restoration of other ISTS lines at Naubatpur is highly desired for enhancing system reliability and security.
- ❑ Matter was discussed in 241st OCC meeting where SLDC Bihar updated that a meeting will be held with PGCIL and BSPTCL in first week of August to resolve the issue. So far, any update in the matter has not been received.

Deliberations in the meeting

SLDC Bihar updated that a meeting was held between BSPTCL and BGCL on 3rd August 2026. The work is planned to be completed by the 1st week of September 2026.

OCC Decision

BSPTCL and SLDC Bihar were advised to coordinate with Powergrid and other stakeholders for ensuring earliest possible restoration of 400kV Patna-Naubatpur-Balia Ckt#1 to its original configuration.

c) Shifting of 400kV Barh-Motihari D/C to pile foundation

- ❑ Outage of the 400 kV Barh–Motihari D/C transmission line was sought by POWERGRID vide email dtd. 11th Jul'26 in view of the precarious condition of Tower Location No. 27, situated on the bank of the Ganga River, necessitating urgent destringing and dismantling of the affected tower. An online meeting was convened on 14th Jul'26 to deliberate the shutdown requirement where POWERGRID requested shutdown of the line for destringing and dismantling of the affected tower and proposed restoration of the line by 20th August 2026. Thereafter, based on site visit report of the committee comprising of POWERGRID, BSPTCL and NTPC Barh, outage of the line was allowed from 21st Jul'26 to 20th Aug'26. POWERGRID was also requested to submit weekly progress reports to ERPC and ERLDC for effective monitoring.
- ❑ Subsequently, POWERGRID updated that the restoration of the line is delayed and expected to be revived by 10th Sep'26 during the online coordination meeting held on 11th Aug'26 for facilitating outage of 400kV KHSTPP-Banka D/C.

Deliberations in the meeting

Powergrid ER-I presented the details of pile foundation works started on bank of Ganga river (from 12.08.26) for restoration of the 400kV Barh-Motihari D/C line.

- ✓ *Continuing river erosion and associated site risks were also highlighted.*
- ✓ *The earlier completion target could not be achieved due to water-logged site and overflowing river conditions after heavy rainfall.*
- ✓ *At present, owing to unfavorable conditions on the river bank, work is stalled temporarily. Present status and site conditions detailed at **Annex B.2.2.c***

OCC Decision

- *Powergrid was advised for **adhering to the timeline (by 10.09.2026)** for restoration of 400kV Barh-Motihari D/C line.*

- Once the barge arrives at the site, Powergrid was advised to intimate to ERPC/ERLDC on the commencement of tower erection works.
- Powergrid shall share progress photographs, detailed activity schedule and regular progress updates with ERPC/ERLDC.

d) Restoration of Koderma ICT #1

The ICT#1 at Koderma (DVC) had been taken out of service since 10th Feb'26 due to a rising trend of acetylene (C₂H₂) concentration in the transformer oil. The ICT was proposed to be sent to the factory for necessary rectification. Meanwhile, one ICT from DSTPS was diverted and installed at Koderma. After commissioning, the diverted ICT was idle charged from the 400 kV side on 02.08.2026. Subsequently, Dissolved Gas Analysis (DGA) of the transformer oil sample indicated a rising trend of acetylene (C₂H₂) concentration. Accordingly, the ICT was taken under shutdown on 04.08.2026 for carrying out further investigation and necessary corrective action.

Deliberations in the meeting

DVC submitted:

ICT 1 was taken under shutdown on 04.08.2026 and ICT 2 was taken under shutdown on 19.08.2026 due to DGA violation. Bushing problem is suspected in both the ICTs and order for detailed internal inspection is placed with M/S Hitachi.

OCC Decision

- DVC was advised to expedite restoration of the both ICT#1 and ICT#2 at Koderma.
- OCC noted that shutdown of 220 kV Maithon(PG) bus will not be availed by Powergrid in September 2026.

2.3 Standardization of Automatic Demand Management Scheme (ADMS) –ER: ERLDC

In 240th OCC meeting, it was decided that a uniform ADMS logic will be formulated for all ER constituents. Accordingly, a meeting was called on 07.07.2026, wherein the uniform philosophy was shared with all ER constituents, and it was decided that it would be implemented by each constituent within 15 days of the meeting. Following action points by the states were agreed upon:

- Confirmation regarding implementation of the standardized ADMS logic (Frequency less than 49.7 Hz for more than 3 minutes delay and OD of more than 150 MW or 12%).
- ADMS testing reports confirming the healthiness of the scheme.
- Communication healthiness reports demonstrating successful execution of ADMS commands up to the end load point.

However, no update has been received from any constituent so far. MoM is attached as **Annexure-2.3**. Logs indicating triggering of ADMS operation and quantum of load shed are expected to be received from SLDCs whenever the triggering conditions are fulfilled. Recently, during the recent low-frequency events on 11.08.2026 and 12.08.2026, when frequency touched around 49.55 Hz, old ADMS logic was satisfied for multiple times for different constituents. However, no ADMS operation information received for any constituent (Except Odisha for one instance only). Analysis is attached as **Annexure-2.3**. Reason for non-operation of ADMS may be shared by the utilities.

Deliberations in the meeting

- ✦ *West Bengal and DVC confirmed implementation of the revised ADMS logic through email to ERLDC. The same was agreed in the meeting convened by ERLDC.*
- ✦ *Odisha informed that discussions with DISCOMs, agreed in the meeting convened by ERLDC, are in progress and the revised logic would be implemented by end of August 2026.*
- ✦ *Bihar and Jharkhand informed that their respective vendors have been communicated regarding the required logic modification and implementation would be completed by end of August 2026.*

OCC Decision:

- *OCC noted implementation of the revised ADMS logic by West Bengal and DVC.*
- *Odisha, Bihar and Jharkhand shall complete implementation of the standardized ADMS logic by 31.08.2026.*

2.4 Requisition for one no. 400/220 kV, 315 MVA Transformer from the Regional Spare Pool available at Durgapur 400 KV substation of PGCIL on loan basis for 400 kV Arambag (WB) Substation: WBSETCL

- ❑ *400 kV Sub Station at Arambag had 04 (four) Nos. 315 MVA, 400/220 kV Power Transformers (PTRs) with total installed capacity of 1260 MVA. The maximum peak demand recorded at Arambag 400 kV Sub-station reached around 800 MVA in June 2026 and further increase in load is anticipated.*
- ❑ *Due to sudden breakdown of one no. 400/220 KV, 315 MVA Power Transformers (Make: TELK | Sl. No.: 140164-2 | YOM: 2011) at Arambag 400 kV Sub Station on 01.05.2026, the existing functional capacity of the substation was reduced to 945 MVA (3 x 315 MVA) only.*
- ❑ *The loss of the faulted transformer has resulted in severe non-compliance with the standard N-1 Contingency Philosophy. Any further tripping or breakdown of an operational 315 MVA PTR may trigger cascaded system tripping, load shedding, or grid instability across the region.*
- ❑ *Though the procurement of 315 MVA, 400/220 KV Transformer is under process but it is apprehended that supply of above rated Transformer may take around 1.5 years to 02 years.*
- ❑ *Considering the situation, this agenda is placed for allotment of one no. 400/220 kV 315 MVA Transformer from the Regional Spare Pool available at Durgapur 400 KV substation of PGCIL on temporary basis for Arambag 400 kV Substation. The same would be returned after supply and commissioning of transformer which is presently under procurement.*

Deliberations in the meeting

WBSETCL submitted:

- *The requirement arose following failure of one of the four 400/220 kV, 315 MVA transformers at Arambagh sub-station (subjected to Y-phase winding fault). The unit could not be restored at site and has been sent to workshop for repair.*
- *Procurement of new transformer has been initiated, with an indicated delivery timeline of around 36 months.*
- *With only three transformers now in service and present loading at around 800 MVA, the sub-station is not maintaining N-1 security. Arambagh was highlighted as an important supply point for the Kolkata (state-capital area) load pocket, adding to the criticality of the situation.*

- It was requested for diversion of a spare transformer (from ER spare pool) available at Durgapur(PG) to Arambagh (WB) S/S on a temporary loan basis till pending repair of the faulty unit or arrival of a new transformer already on order.

WB SLDC substantiated that during upcoming Durga Puja festival, N-1 condition will not be maintained at Arambagh(WB) with three existing transformers

Powergrid stated that diversion of the spare ICT will require approval from the Corporate HQ as well as highlighting availability of only one spare ICT (315 MVA) in ER-II at present, i.e. at Durgapur(PG). This ICT, though old (1991), has been refurbished with new bushings and sound healthiness is assessed. Another spare 315 MVA ICT is available at Jamshedpur(PG) posing constraints in inter-state transportation.

OCC Decision

- After detailed deliberation and to meet demand during upcoming Durga Festival, OCC forum granted in-principle consent to the diversion of a spare 400/220 kV transformer from Durgapur(PG) to Arambagh on loan basis and referred to TCC/ERPC for concurrence.
- WBSETCL was advised to expedite its ongoing procurement of the replacement transformer and to pursue repair of the faulty unit in parallel, so that the diverted spare can be returned at the earliest.
- **All ER states were also advised to maintain spare pool of 400/220 kV transformers to deal with intra-state network contingencies. They were advised to submit details of their in-service ICTs, spare ICTs and ICTs under procurement (along with tentative delivery schedule) at present in next OCC meeting.**

2.5 Reconductoring of 400kV TSTPP-Meramundali D/C lines: ERLDC

- ❑ Reconductoring works of 400kV TSTPP-Meramundali D/c lines was approved under ERES-43 with completion schedule of 02.03.2026. The reconductoring work in the Ckt#2 LILO part was started from 20.11.2025 with a target to complete LILO part within one month by 20.12.2025. However, only 15Ckm out of 35Ckm in LILO portion could be completed till Dec'25. Owing to the very slow progress, reconductoring in LILO part was paused and reconductoring in Ckt#1 was started from 06.01.2026. Subsequently, the progress of the work and network scenarios were reviewed in a special meeting held on 06.03.2026 participated by ERLDC, ERPC, SRPC, SRLDC, SLDC Odisha and POWERGRID where it was decided not to allow shutdown of 400kV TSTPP-Meramundali #1 reconductoring works with immediate effect, in view of prevailing network conditions in ER and high SR demand. It was also decided that further reconductoring will be continued after reduction of peak Demand period of Bhubaneswar i.e. after July 2026.
- ❑ To expedite the reconductoring work of both the circuits, it was decided that LILO portion of the ckt 2 will be isolated by bypassing the ckt at LILO point and make it available for reconductoring of the remaining part, after arrival of Monsoon, when demand of Odisha / Bhubaneswar will be comparatively lower & due to lower ambient temperature compared to summer, the thermal rating of the line will increase, and hence the threshold current setting for SPS operation can be increased. As per the above decision, POWERGRID was advised to plan accordingly.
- ❑ Post summer, an online meeting on dtd. 06.07.2026 was held with representatives of ERPC, ERLDC, POWERGRID, TSTPP and SLDC Odisha to discuss outage plan of reconductoring work and NH diversion proposal of HVDC Talcher-Kolar bi-pole. In the meeting, POWERGRID was advised to assess the site conditions for starting the reconductoring work in the LILO portion of Ckt#2. After site assessment by POWERGRID, another online meeting was held on

13.07.2026. POWERGRID intimated plan to resume the reconductoring work by 3rd / 4th week of July 2026 depending on manpower mobilization.

- In the 241st OCC meeting, POWERGRID informed that deployment of agency for the reconductoring/ bypass work has been delayed, and the agency is expected to be mobilised by end-July'26. It was decided to isolate the LILO portion and start reconductoring work by end of Jul'26 without any further delay. Subsequently, the LILO part of 400kV TSTPP-Meramundali #2 was made available for reconductoring on 07.08.2026 after switching off the line for a brief period and isolating the LILO part.
- Thereafter, a special coordination meeting was called by SRPC on 05.08.2026 involving representatives from ERPC, ERLDC, SRLDC, TSTPP and POWERGRID to finalize TSTPP unit outage schedule in coordination with proposed HVDC Talcher-Kolar bi-pole shutdown during November. In the said meeting, outage plan for reconductoring of 400kV TSTPP-Meramundali D/C was also discussed. It was agreed that reconductoring of LILO portion is to be completed at the earliest. Subsequently, reconductoring of balance 9 Ckm in 400kV TSTPP-Meramundali Ckt#1 to be completed before planned outage of HVDC Talcher-Kolar scheduled in November.
- Subsequently, outage of 400kV TSTPP-Meramundali #2 was allowed on 07.08.2026 to isolate the LILO part for resuming the reconductoring work.

Deliberations in the meeting

- ✓ *POWERGRID Odisha informed that shutdown of the LILO portion was availed from 08.08.2026; however, only about 1.2 circuit-km of reconductoring could be completed due to repeated rainfall.*
- ✓ *POWERGRID agreed to avail shutdown of 400 kV Talcher–Meramundali Ckt-1 from 01.09.2026 for carrying out the remaining approximately 9 km reconductoring work.*
- ✓ *It was informed that bay upgradation requires at least 60 days and could not be suitably accommodated before completion of Ckt-1 reconductoring.*
- ✓ *The planned shutdown of ± 500 kV Talcher–Kolar HVDC from 01.11.2026 and the coinciding Talcher TPS Unit-5 shutdown were also considered while sequencing the works.*

OCC Decision

Shutdown for Circuit-2 reconductoring will be considered only after Circuit 1 and the LILO portion are fully completed and re-energised, to safeguard supply reliability during the work. Accordingly, shutdowns in the Talcher–Meramundali/Talcher–Kolar corridor shall be availed in the following chronological sequence:

- 1. LILO portion of Talcher–Meramundali Ckt-2: 08.08.2026 to 31.10.2026*
- 2. 400 kV Talcher–Meramundali Ckt-1: 01.09.2026 to 31.10.2026(after ckt-2 is taken into service on completion of reconductoring work in the LILO section.)*
- 3. ± 500 kV Talcher–Kolar HVDC: 01.11.2026 to 15.11.2026*
- 4. 400 kV Talcher–Meramundali Ckt-2: from 16.11.2026 – till mid Feb '27.(after restoration of Talcher-Kolar HVDC bipole)*

The timeline for the associated bay-upgradation work may be finalized in a separate special meeting.

2.6 Shutdown of 400 kV Chandauti (PMTL)–Nabinagar (NPGC) D/C Lines for NH work: ERLDC

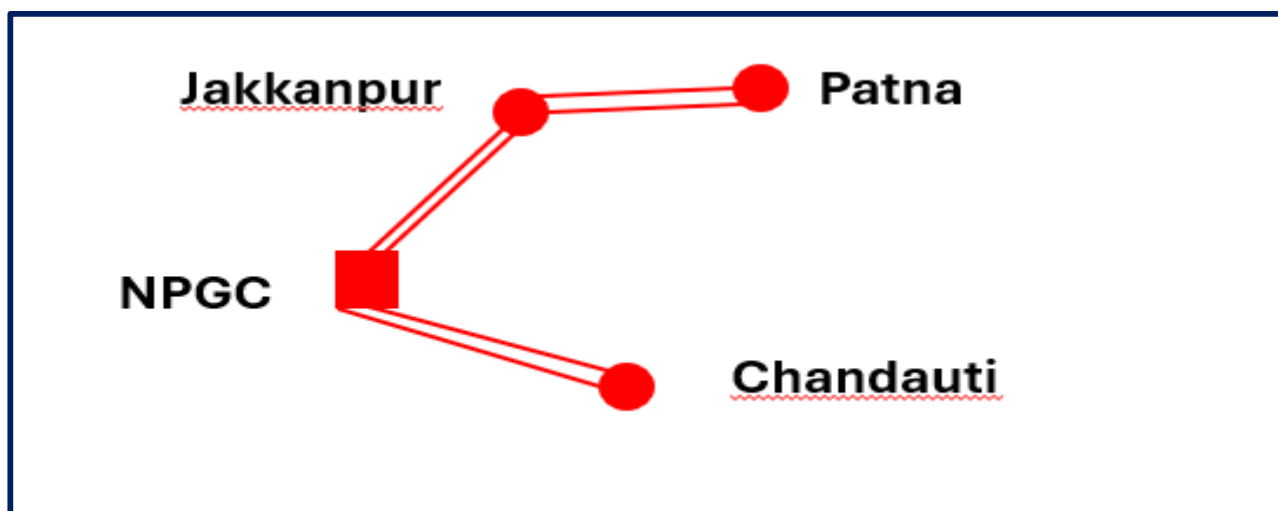
Construction of the Four-Lane Access-Controlled Greenfield National Highway under the Bharatmala Pariyojana in the State of Bihar requires shifting/raising of the 400 kV Chandauti (PMTL)–Nabinagar (NPGC) D/C transmission lines. Accordingly, POWERGRID received a request from NHA for shutdown of both circuits of the 400 kV Chandauti (PMTL)–Nabinagar (NPGC) D/C lines to facilitate the proposed work.

Initially, shutdown was requested from 01.08.2026 to 20.08.2026 (20 days). Nabinagar (NPGC), an NTPC generating station with an installed capacity of 3×660 MW, is presently associated with the following 400 kV transmission corridors for evacuation of entire generations:

- 400 kV Nabinagar (NPGC)–Chandauti (PMTL) D/C
- 400 kV Nabinagar (NPGC)–Jakkanpur D/C

Both lines are Quad Moose conductor configurations with a transformation/transfer capability of approximately 1748MVA (1700 MW).

The schematic diagram of the associated 400 kV network of NTPC Nabinagar (NPGC) is as follows:



- From the network connectivity, it is observed that during the proposed shutdown of both circuits of 400 kV Chandauti (PMTL)–Nabinagar (NPGC) D/C line, only 400 kV Nabinagar (NPGC)–Jakkanpur D/C line would remain in service. Consequently, the entire generation of NTPC Nabinagar (NPGC), approximately 1800–2000 MW, would be dependent on a single transmission corridor, thereby posing a significant risk of total generation loss in the event of any further contingency/tripping during the shutdown period.
- The matter was deliberated in the 241st OCC Meeting, wherein ERLDC highlighted that shutdown of 400 kV Chandauti–NPGC D/C lines would necessitate generation restriction up to 1500 MW at NPGC out of total generation of 1800-1900MW to maintain transient stability of the system. Accordingly, it was recommended that the proposed shutdown may preferably be availed during the planned annual overhauling of any unit of NPGC, so that the associated system risk can be minimized. The same was also deliberated in the 241st OCC Meeting held on 17.07.2026 at ERPC Secretariat, Kolkata.
- However, during the deliberations, the possibility of temporary restoration of the line through deployment of Emergency Restoration System (ERS) was also discussed.

- ❑ Subsequently, POWERGRID, vide email dated 08.08.2026, resubmitted the request letter from NHAI, intimating the revised work execution plan for availing shutdown of the 400 kV Chandauti–NPGC D/C lines from 16.08.2026 to 11.09.2026. The request was accompanied by a joint site-visit report, indicating the availability of required manpower and materials, as well as favourable site conditions for commencement of the work.
- ❑ In view of the above, the proposed shutdown involves significant operational and system-security implications, particularly considering the dependence of NPGC generation on a single remaining transmission corridor during the outage.

Deliberations in the meeting

ERLDC highlighted the following:

- ✓ *With outage of 400kV NPGC-Chandauti D/C, only 2 nos of evacuation path i.e. 400kV NPGC-Jakkanpur D/C will remain for evacuation of full generation of 3 nos of 660MW units.*
- ✓ *In case of any N-1 contingency, there will be issue of dynamic stability in machine for evacuation of full generation and may lead to generation loss. To avoid this, it may necessitate restricting NPGC generation to around 1500 MW to ensure dynamic stability under (n-1) line contingency.*
- ✓ *Accordingly, it was proposed that the diversion work may be taken up in conjunction with planned Annual Overhauling of an NPGC unit.*

SLDC Bihar apprised that around 1500 MW share of Bihar is fed from NPGC, thus, considering ongoing peak demand season in Bihar, the proposed S/D may be availed in November'26 during outage of NPGC units.

NHAI stressed urgency of the shutdown in view of criticality of the highway project.

OCC Decision

- ✓ *Keeping in view of the power evacuation constraints, OCC forum agreed to the S/D of 400 kV Chandauti–NPGC D/C lines from 25th November 2026 for the requested duration i.e 26 days.*
- ✓ *Powergrid ER-I was advised to convey the agreed shutdown schedule to NHAI for necessary arrangements.*
- ✓ *POWERGRID/PMTL shall submit the revised shutdown programme, route/crossing diagram, execution duration and contingency plan before availing the shutdown.*

2.7 Agenda by NTPC Kahalgaon: NTPC

- ❑ **132 kV Kahalgaon–Lalmatia Line outages are being approved without obtaining the consent of NTPC Kahalgaon**, despite the line being of critical importance from the power generation perspective. During the recent outages of the **132 kV Kahalgaon–Lalmatia Line**, coal supply from the **Lalmatia Mines** to NTPC Kahalgaon was severely affected. The outages resulted in **low voltage conditions at the Lalmatia end**, adversely impacting mining operations and significantly reducing coal production and dispatch to the station. Since the uninterrupted availability of this line is vital for maintaining reliable power supply to the mining operations and ensuring continuous coal availability to NTPC Kahalgaon, it is requested that **no outage of the 132 kV Kahalgaon–Lalmatia Line be approved without prior consultation of NTPC Kahalgaon**, except under emergency conditions.
- ❑ Implementation of PLCC/OPGW-based communication and protection for the three 132 kV transmission lines from NTPC Kahalgaon: Kahalgaon–Sabour, Kahalgaon–Lalmatia, and Kahalgaon-Kahalgaon. As per CEA Grid standard fault in 132kV system should be cleared in 160 msecs, most of the times Kahalgaon substation end breaker fails to OPEN during line faults and the fault gets cleared only by opening of breaker at NTPC Kahalgaon.

- A **three-day shutdown of 200 MVA ICT#1 (400/132 kV)** has been proposed for the replacement of the **132 kV (LV side) circuit breaker**. Despite repeated follow-ups, the outage approval has not yet been issued by the **ERLDC Outage Team**, as the shutdown has not been concurred by the **Bihar SLDC and Jharkhand SLDC**. The continued deferment of this critical maintenance activity may adversely affect the **reliability and operational security of the 132 kV system at NTPC Kahalgaon**. Early approval of the proposed shutdown is, therefore, requested to facilitate timely replacement of the old breaker and ensure reliable system operation.
- As communicated earlier during the OCC meeting, the **400 kV Sectionalizer-1 and Sectionalizer-2 breakers at NTPC Kahalgaon are presently being kept in the CLOSED condition**, considering the high power flow requirement during the summer months. In view of the prevailing system conditions, **clarity may kindly be provided regarding the continued operation of the 400 kV Sectionalizer-1 and Sectionalizer-2 breakers in the CLOSED condition** or whether any change in the present operating philosophy is envisaged.

Deliberations in the meeting:

- *NTPC Kahalgaon raised concern that a 132 kV line outage (Kahalgaon–Almaty/associated feeder) had been approved without obtaining NTPC's concurrence as the affected generating station; NTPC requested that generating stations be included as concerned parties in future OCC approvals affecting their evacuation network.*
- *It was reported that the earlier line outage caused a voltage dip on NTPC's internal 33 kV system, leading to tripping of drives/auxiliaries at the plant; this recurred on a subsequent occasion for the same reason.*
- *ICT-1 (132 kV side) breaker has a hydraulic issue requiring an approximately 3-day shutdown; approval for the associated outage was complicated by a mismatch between neighbouring SLDCs — Bihar SLDC had approved the outage while Jharkhand SLDC had not, delaying execution.*
- *A 400 kV bus sectionalizer, made operational in April, was subsequently kept in closed condition through the summer months on advisory grounds (to support higher power-flow requirements); NTPC sought clarity on when normal (open) configuration would be restored.*
- *A reliability concern was raised regarding instances where a sub-station breaker fails to open on a fault and the fault is instead cleared by a different (delayed/back-up) protection stage, resulting in extended fault-clearing time; NTPC requested review of the protection time-delay settings.*
- *NTPC proposed implementation of PLCC/OPGW-based tele-protection communication for certain outgoing 132 kV lines, citing 5–6 tripping events since 28th March attributable to extended fault-clearance duration on those lines leading to unit tripping; this proposal was stated to have been submitted previously and NTPC sought its implementation.*

OCC Decision

- *OCC advised all Generating Stations including NTPC to participate in OCC outage meetings regularly for cognizance of any shutdown/outage approvals affecting their respective stations network. The concerned SLDCs (Bihar and Jharkhand) were advised to properly coordinate and align their approvals with NTPC.*
- *Restoration of the 400 kV bus sectionalizer to normal (open) configuration to be reviewed once the high-power-flow (summer) requirement eases.*

2.8 Reliability and availability of transmission lines connected to NTPC Barh: NTPC

- NTPC Barh is presently having eight transmission lines for evacuation of generated power. However, Barh-Motihari-1 and Barh-Motihari-2 transmission lines are currently under outage till 10 September 2026 due to tower relocation activities. Consequently, only six transmission lines are available for power evacuation from the station.
- It is observed that since 23 June 2026, a total of 16 disturbances has occurred in the transmission system associated with NTPC Barh. Details are as follows :-

Details of Line Disturbances Recorded at Barh End				
S/N	Date & Time	Details of 400kV Lines	Details of Fault	Remarks
1	07.08.2026, 13:55 hrs	Barh-Bakhtiyarpur-1	Line tripped from remote end but remained charged from Barh end	Line remained charged from Barh end
2	06.08.2026, 10:04hrs	Barh-Bakhtiyarpur-1	During S/D, DT not received at Barh after opening CB from remote	CB opened by manual command from Barh end
3	05.08.2026, 11:53hrs	Barh-Patna-3	Protection picked up, RΦ, 1.636kA	Line remained charged
4	05.08.2026, 11:53hrs	Barh-Patna-4	Protection picked up, RΦ, 1.702kA	Line remained charged
5	05.08.2026, 07:45hrs	Barh-Patna-3	Protection picked up, RΦ, 1.86kA	Line remained charged
6	05.08.2026, 07:45hrs	Barh-Patna-4	Protection picked up, RΦ, 1.952kA	Line remained charged
7	21.07.2026, 15:05hrs	Barh-Patna-3	Zone-1,RΦ, 18.64kA	Line Tripped
8	20.07.2026; 05:26 hrs	Barh-Kahalgaoon-2	Zone-1; YΦ, 3.73kA, 136.1km	A/R successful at Barh but failed at remote end
9	20.07.2026; 05:45 hrs	Barh-Kahalgaoon-1	Zone-1; BΦ, 7.99kA, 50.25km	A/R successful; Line remained charged
10	18.07.2026; 01:27 hrs	Barh-Patna-3	Zone-1; BΦ, 10.28kA, 25.77km	A/R successful; Line remained charged
11	16.07.2026, 14:12 hrs	Barh-Bakhtiyarpur-2	During S/D, DT not received at Barh after opening CB from remote	CB opened by manual command from Barh end
12	15.07.2026; 18:18 hrs	Barh-Bakhtiyarpur-1	Zone-2&3 protection picked up, YΦ, 6.11kA	Line remained charged
13	15.07.2026; 18:18 hrs	Barh-Patna-4	Protection picked up, YΦ, 1.21kA	Line remained charged
14	26.06.2026; 03:54 hrs	Barh-Motihari-1	Protection picked up, RΦ-972A and BΦ-831.2A	Line remained charged
15	26.06.2026; 03:54 hrs	Barh-Motihari-2	Protection picked up, RΦ-1.184kA and BΦ-952.9A	Line remained charged
16	23.06.2026; 00:38 hrs	Barh-Patna-3	Zone-3; YΦ, 2.42kA, 112.3km	A/R successful; Line remained charged

- ☐ In view of the reduced transmission network availability and recurring line disturbances, ensuring reliable power evacuation from NTPC Barh is becoming critical.

- ❑ A recent instance highlighting above concern occurred on **07 August 2026**, when four generating units of NTPC Barh were in service. At **11:14 hrs**, an emergency shutdown of **Barh-Patna-4** transmission line was taken due to an R-phase hotspot observed at the Patna end. Subsequently, at **13:55 hrs**, **Barh-Bakhtiyarpur-1** transmission line tripped owing to problem at the Bakhtiyarpur end. As a result, the number of available transmission lines were reduced to four, leaving the station in a vulnerable transmission configuration for power evacuation.
- ❑ Although both transmission lines were restored before the evening peak demand period, any additional transmission line outage or prolonged restoration of Barh-Patna-4 and Barh-Bakhtiyarpur-1 could have necessitated generation backing down at NTPC Barh during peak hours.
- ❑ Moreover, Kahalgaon Banka transmission line is proposed for emergency shutdown for 9 days starting 13th August 2026 which will further aggravate power evacuation situation at NTPC Barh due to power import from Kahalgaon to Barh. It is requested that necessary measures may kindly be taken to enhance the reliability and availability of the transmission network associated with NTPC Barh particularly during the period when the Motihari transmission lines remain under outage.

Deliberations in the meeting

NTPC, Barh raised the following concerns:

- ✓ *In view of the 16 disturbances observed in transmission lines associated with NTPC Barh since 23.06.2026, it was proposed that a dedicated coordination team, along with a dedicated WhatsApp group, be constituted to facilitate prompt coordination and restoration in the event of any transmission line outage. In addition to the ERLDC Control Room, contact details of designated responsible personnel from ERLDC, POWERGRID and BSPTCL may be shared to ensure round-the-clock communication during emergencies.*
- ✓ *In the event of any transmission constraint affecting the three interconnected generating stations, namely NTPC Barh, NTPC Kahalgaon and NTPC Farakka, the resultant impact on the loading of the available transmission corridors/lines may be assessed by ERLDC. An appropriate SOP covering the operational actions to be taken under such scenarios may accordingly be shared by ERLDC.*
- ✓ *To prevent overloading of any transmission line beyond 1100 MW, NTPC Barh would immediately back down generation in consultation with ERLDC.*

OCC Decision

- *POWERGRID was advised for ensuring early restoration of 400 kV Kahalgaon– Banka D/C & 400kV Barh-Motihari D/C lines.*
- *POWERGRID was also advised to direct its internal task force to proactively inspect the entire line corridor and ensure the healthiness and reliability of all associated transmission elements, particularly before and throughout the shutdown period of the 400 kV Kahalgaon– Banka D/C, so as to ensure smooth and reliable evacuation of generation from Barh TPS under the prevailing network condition.*

2.9 Reporting of Monthly Commissioned/De-commissioned Generation Capacity in the State Control Area: ERLDC

- ❑ Grid-India is in the process of establishing a structured and uniform mechanism across all regions for the collection of information regarding generation capacity commissioned or de-commissioned within the respective State control areas. The objective is to facilitate the timely compilation of comprehensive monthly commissioning and de-commissioning statistics at the All-India level.

- Accordingly, all SLDCs/State Utilities are requested to furnish the details of FTC and COD of generation systems, along with the details of any de-commissioning, as applicable, in the format provided in the enclosed Annexure.
- The information furnished by the SLDCs/State Utilities will facilitate the timely compilation of comprehensive monthly statistics relating to generation capacity additions and de-commissioning at the All-India level.

Deliberations in the meeting

- The forum acknowledged the importance of maintaining a proper and regular record of generation capacity commissioned and de-commissioned within the State control areas.
- OCC noted and requested all ER SLDCs to furnish the above information in the enclosed format(**Annex B.2.9**) on monthly basis within the stipulated timeline, i.e., by the 7th day of every succeeding month, for timely compilation of the data at All-India level.
- It was agreed that ERLDC would provide a Google Sheet for submission and regular updation of the required information. SLDCs may update the details of generation capacity commissioned or de-commissioned during the previous month in the designated sheet by the 7th day of every succeeding month.

2.10 Shutdown proposal of Thermal generating units: ERPC

- ✓ **Rescheduling of Start of Overhauling of Unit-5 of NTPC Kahalgaon: NTPC**
- In place of the planned overhauling of **Unit-5 (500 MW)** of Stage-II, NTPC Kahalgaon, which was scheduled to commence from **31.8.2026** to **05.10.2026**, **Unit-6 (500MW)** of Stage-II is scheduled for planned overhauling from **10.09.2026** to **09.10.2026..**

Planned shutdown: MPL

MPL Unit #2 (Maithon) is scheduled from **20th November 2026 to 4th January 2027** as per the approved outage schedule.

During this period, the Turbine OEM experts are scheduled to visit the plant for carrying out turbine inspection and maintenance activities. In parallel, the **Boiler Annual Overhaul (AOH)** activities will also be undertaken.

Deliberations in the meeting

- ✓ NTPC proposed a 35-day shutdown for Unit 6 (Kahalgaon STPP) from 10th September 2026 in place of Unit#5. It was also updated that U#3 which is under planned outage since 28th July 2026 shall be restored by 10th September 2026.
- ✓ The beneficiary utility (Bihar) confirmed no objection to the proposed shutdown, given its relatively small share in the affected station's capacity.

OCC Decision

OCC forum agreed with the following planned overhauling shutdowns:

- NTPC Kahalgaon Unit#6 from 10.09.2026 to 09.10.2026.
- MPL Unit #2 from 20th November 2026 to 4th January 2027 for Turbine inspection and Boiler overhaul activities.

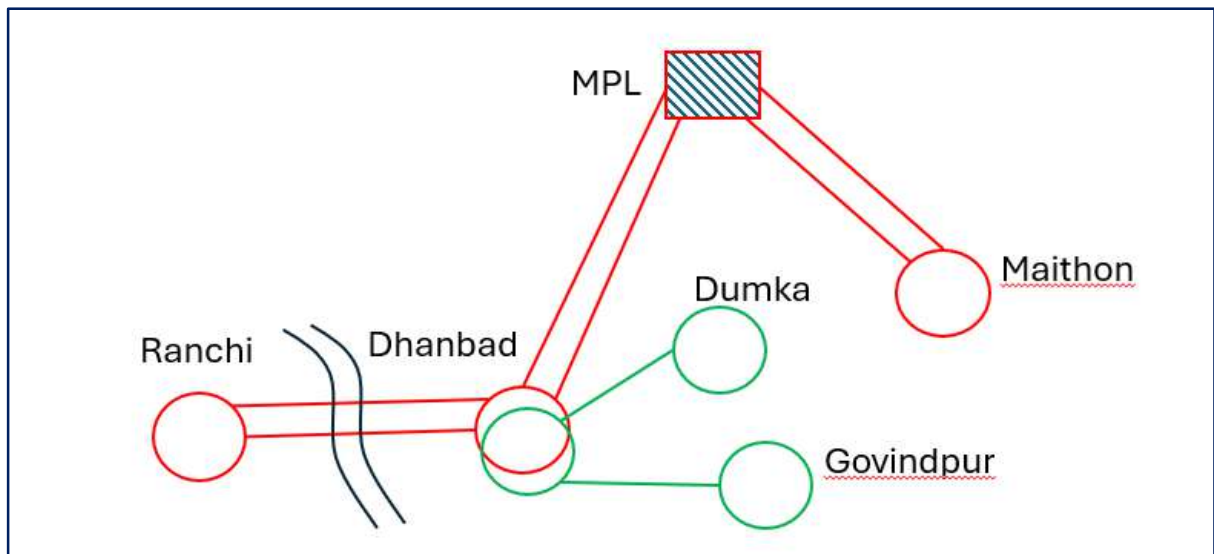
ADDITIONAL AGENDA

2.11 Shutdown proposal of 400kV Ranchi-Dhanbad D/C in Sep'26 for NHA1 diversion work: ERLDC

Request has been received for continuous shutdown of 400kV Ranchi-Dhanbad D/C from 01-Sep-2026 till 11-Sep-2026 for NHA1 diversion work. The said transmission line serves as a vital link for evacuation of power generation from MPL, an IPP with an installed capacity of 2×525 MW which is presently having the following 400 kV transmission lines for generation evacuation:

- 400kV MPL-Maithon(PG) D/C line (Twin HTLS)
- 400kV MPL-Dhanbad(NKTL)-Ranchi D/C line (Twin Moose)
- 2x 400/220kV 500MVA ICT at Dhanbad (NKTL)

The schematic diagram of the associated 400 kV network of MPL is as follows:



During the proposed shutdown of 400kV Ranchi-Dhanbad D/C, evacuation corridor of MPL will be limited to

- 400kV MPL-Maithon(PG) D/C line (Twin HTLS)
- 2x 400/220kV 500MVA ICT at Dhanbad (NKTL) (Connected with only 2 220

In case of any forced outage of any circuit of 400kV MPL-Maithon, the associated generation of MPL will be one tripping (another MPL-Maithon Ckt) away from total outage as 2 nos. downstream 220kV lines from Dhanbad(NKTL) will not be able to evacuate the generated power of MPL.

It is proposed that the shutdown of the line may be allowed in November during planned AOH of MPL unit#2 to limit generation loss in case of any contingency.

Deliberations in the meeting

ERLDC submitted:

- ✓ Shutdown of the MPL (Maithon-area) double-circuit connectivity line was proposed for 1st September, for a duration of about 11–12 days, linked to NHA1-related work.
- ✓ This was flagged as a particularly high-risk or critical shutdown since with the double circuit out, MPL's generation (2×525 MW) would need to evacuate entirely through the remaining downstream 220 kV network; any contingency (e.g., a hotspot-related trip) on those downstream lines during the

outage window could force tripping of the entire MPL generation and may end up damaging the 220 kV network.

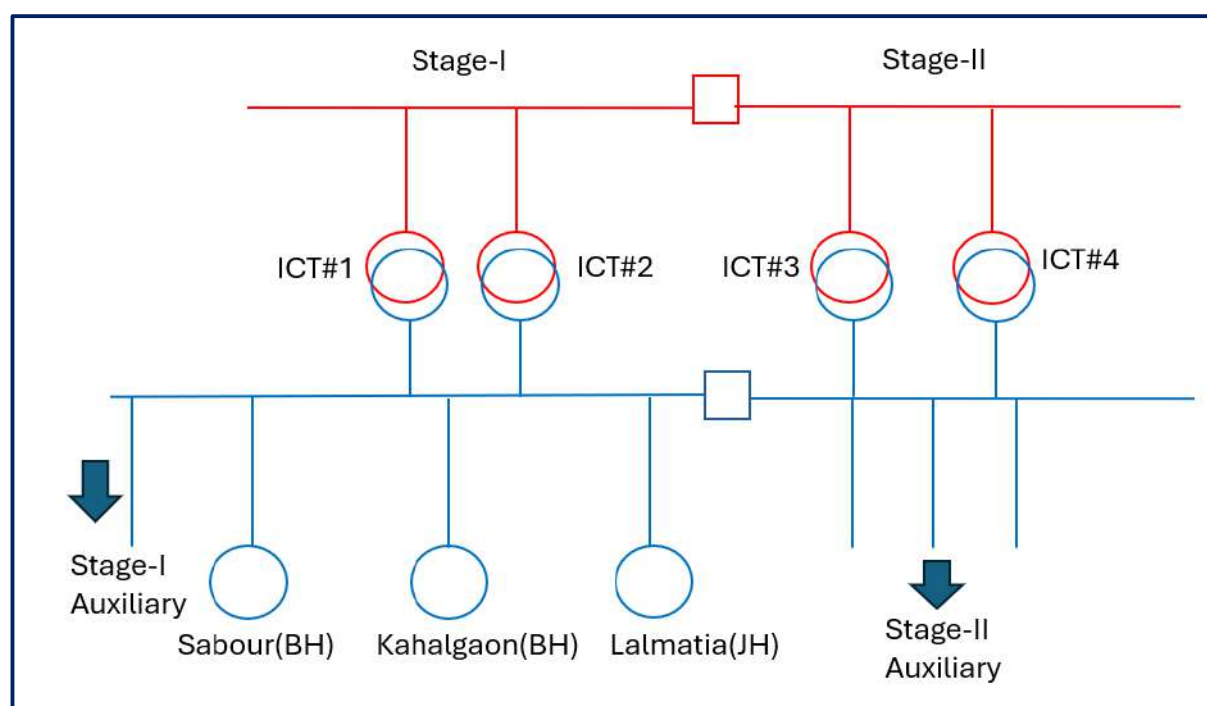
- ✓ Appropriate operational restrictions (e.g., on ICT loading, or a defined response protocol at Dhanbad-end ICTs in case of a contingency) would be needed to safely accommodate the shutdown, given the criticality.
- ✓ An alternative was discussed of deferring this shutdown to November, to align with other already-planned unit outages (e.g., a unit outage schedule at a related station in November) and reduce concurrent-risk exposure.

OCC Decision

OCC opined to convene a special meeting comprising MPL, DVC, Powergrid and NHAI to finalize the shutdown schedule.

2.12 Higher loading in 400/132 kV 200MVA ICTs at NTPC Kahalgaon: ERLDC

The loading of 400/132kV ICTs at NTPC Kahalgaon remained on relatively higher side post shutdown of 400kV KHSTPP-Banka D/C S/D as reported by NTPC KHSTPP control room. As the 132kV buses downstream of ICT#1,2 (2x200MVA) (Stage-I) and ICT#3,4 (2x200MVA) (Stage-II) are not coupled, there is uneven loading of the ICTs of stage-I and stage-II. While ICT#1&2 remain loaded up to 100MW each, loading in ICT#3&4 remain below 50MW combinedly. For enhanced utilization of the 400/132kV ICTs, NTPC was requested to explore feasibility of coupling 132kV buses. The schematic arrangement of the network is presented below:



Besides, 132kV KHSTPP-Sabour line has been upgraded to HTLS and charged on 08-Aug-2026. However, O/C protection settings i.r.o. 132kV KHSTPP-Sabour(BH) line are kept at 550A (~125MW) at KHSTPP end to limit loading of the 400/132kV ICT#1,2 whereas the line is capable of operating at 900A(~205MW) post reconductoring of line.

However, NTPC is not agreed to couple 132kV buses citing reliability concerns.

Deliberations in the meeting

ERLDC submitted :

- ✓ *It was informed that on completion of augmentation, the loading capability of ICT-1 and ICT-2 (downstream) would go up to 100 MW per transformer.*
- ✓ *Once ICT-3 and ICT-4 (new bays) are commissioned and associated inter-connections established, the existing high-loading condition on ICT-1 and ICT-2 is expected to ease, as load can be shared across the additional transformers.*
- ✓ *A protection-related concern was flagged in respect of one of the loading arrangements, requiring further data/clarification from the utility.*
- ✓ *This is fundamentally a system-operation trade-off (loading relief vs. fault-containment) that needs to be decided from an overall grid-reliability perspective rather than purely a station-level preference*

NTPC submitted :

The two 132 kV systems (old bay associated with ICT-1 & 2 and new bay associated with ICT-3 & 4) are physically separated by approximately 1 km; a scheme for inter-connection using the existing 132 kV cable/feeder was discussed, including provision to keep the link normally open and close it only as a fault-restoration or backup arrangement, subject to cable rating and protection coordination.

OCC Decision

NTPC was advised to explore the replacement of old inter-connecting 132 kV cable and accordingly intimate ERPC/ERLDC at the earliest.

2.13 Long outage of 400/220kV 315MVA ICT#1 at NTPC Talcher: ERLDC

- ☐ The 400/220kV 315MVA ICT#1 at TSTPP has been under long outage since 01-Nov-2024 due to operation of PRD. As per latest email received from TSTPP on dtd. 13-Aug-2026, the ICT is presently undergoing rewinding at M/s CGPISL Bhopal and it is expected to be in service by mid of Nov'26.
- ☐ The availability of both the ICTs at TSTPP are important considering proposed shutdowns of HVDC Talcher-Kolar bi-pole planned from 1st week of November to limit restrictions on TSTPP generation. Besides, it also helps to maintain reliability and serve enhanced evacuation path during planned long shutdown of 400kV TSTPP-Meramundali Ckt#1 and Ckt#2.
- ☐ NTPC should give all out efforts to expedite the restoration and bring the ICTs to service by last week of Oct'2026 before the planned outage of HVDC Talcher-Kolar bi-pole.

Deliberations in the meeting

ERLDC highlighted the constraints in system reliability amid proposed shutdowns of HVDC Talcher-Kolar bi-pole planned from 1st week of November and need for enhanced evacuation path during planned long shutdown of 400kV TSTPP-Meramundali Ckt#1 and Ckt#2.

OCC Decision

NTPC was advised to restore the 315MVA ICT#1 at TSTPP positively by end of Oct'2026.

2.14 Technical Relaxation granted to NTPC Barh Stage-I: NTPC

CEA has granted technical relaxation to NTPC Barh Stage-I for three years with respect to MTL Level and Ramp rate.

Quote:-

1. *Relaxation for Minimum Power Level to 80% - [Regulation 6 (Minimum power level capabilities of coal based thermal power generating units for flexible operation)]*
2. *Relaxation in the ramp rate to 0.1% of capacity corresponding to MCR per minute [Regulation 7 (Ramp rate capabilities of coal based thermal power generating units for flexible operation).*

Unquote

1. The minimum operating power level (turndown level) for NTPC Barh Stage-I has to be revised to 80% of the unit's rated capacity, as against the 55% minimum loading requirement stipulated under IEGC-2023. This relaxation acknowledges the unit-specific operational and technical constraints and permits operation at a higher minimum load level.
2. NTPC Barh Stage-I has been granted relaxation in the ramp rate declaration requirement, allowing a ramp rate of 0.1% of rated capacity per minute in place of the 1% ramp rate prescribed under prevailing regulatory provisions. This relaxation takes into account the station's operational characteristics and provides flexibility in load variation during grid operation.

Deliberations in the meeting

- ✓ *NTPC presented the technical constraints associated with Barh Stage-I(**Annex B.2.14**) and shared the letter issued by CEA for relaxation of norms to improve the reliability. CEA has provided the relaxation on following :*
 - ♦ *MTL of station as 80%*
 - ♦ *Ramp rate as 0.1%.*
- ✓ *The relaxation would be for **03 years** and by this time NTPC has to fix the reliability issues of the station.*
- ✓ *ERLDC was requested to incorporate the MTL and ramp rate in the Scheduling software at the earliest so that the reliability improvement jobs can be started by the station for submitting the quarterly report to CEA.*
- ✓ *Bihar has the major share allocation followed by Odisha & Jharkhand.*

Bihar and Odisha SLDCs raised concerns regarding the commercial impact, noting that continued part-loading below the design-safe level makes the station's power among the costliest in the Eastern Region, creating a cost burden for beneficiaries.

ERLDC apprised that it being a sub-judice matter, the necessary changes will be implemented in the scheduling system after relevant order is received from CERC. It was further apprised:

- ✓ The issue regarding the reduced ramp rate of Barh Stage-I had been discussed in the 220th OCC Meeting of ERPC held on 28th October 2024 and subsequently in the 232nd OCC Meeting of ERPC held on 24th October 2025. During the discussions, NTPC had informed the forum that a petition had been filed before the Hon'ble CERC seeking approval for the proposed relaxations.
- ✓ Provisions of Regulation 45.12 of the IEGC, 2023, which state as follows:
"Minimum turndown level for regional entity thermal generating stations: The minimum turndown level for operation in respect of a unit of a regional entity thermal generating station shall be 55% of the MCR of the said unit or such other minimum power level as specified in the CEA (Flexible Operation of coal based Thermal Generating Units) Regulations, 2023, as amended from time to time, whichever is lower:
Provided that the Commission may fix through an order a different minimum turndown level of operation in respect of specific unit(s) of a regional entity thermal generating station:..."
- ✓ The minimum turndown level of NTPC Barh Stage-I may be considered as 80% instead of the present 55% only after an appropriate order is issued by the Hon'ble CERC. Until such an order is issued by the Hon'ble Commission, the minimum turndown level of NTPC Barh Stage-I may continue to be considered as per Regulation 45.12 of IEGC, 2023, i.e. 55%, for all applicable purposes, including scheduling, SCED, TRAS, SCUC, compensation calculations, etc.

OCC Decision:

- Forum advised NTPC to approach CERC at the earliest so that commercial aspects associated with the relaxed norms may be dealt as per CERC order. Forum further opined that they can support 80% MTL as per relaxation given by CEA to Barh -I .
- NTPC was advised to continue reporting progress on the underlying remedial modification works so that normal Technical Minimum and ramp-rate parameters can be restored once the reliability constraint is resolved.

2.15 Urgent Commissioning of ULDC-III SCADA System before 1st October 2026: SLDC, WB

- ❑ The existing Annual Maintenance Contract (AMC) for the ULDC-II SCADA and associated systems under the jurisdiction of SLDC, West Bengal, entrusted to M/s Chemtrol, is scheduled to expire on 30 September 2026.
- ❑ Considering that M/s Chemtrol has quoted approximately three times the cost of the previous AMC and in view of the recent advisories/guidelines issued by the Government of West Bengal, extension of the existing AMC is not feasible.
- ❑ It is, therefore, imperative to operationalize the ULDC-III SCADA system, with complete system integration and establishment of ICCP connectivity between SLDC, West Bengal and ERLDC(so that all tie line data should be captured to ensure proper drawal data etc), latest by 1 October 2026. After expiry of the existing AMC, M/s Chemtrol will no longer attend to any suspected, erroneous or non-reporting data in the ULDC-II system.
- ❑ Any delay in commissioning ULDC-III may seriously affect real-time grid operation from SLDC WB and could also render the commercial operations of ALDC, WBSEDCL, unmanageable.
- ❑ In view of the criticality and urgency involved, the matter is placed before the OCC forum for sensitizing all concerned departments and organizations. A time-bound roadmap and implementation plan may urgently be finalized to ensure complete commissioning and operationalization of ULDC-III before October 2026.

Deliberations in the meeting

WB SLDC submitted:

- The existing Operation & Maintenance AMC for the ULDC-II system (with the incumbent contractor) with M/s Chemtrol expires on 30th September 2026. This AMC has already been extended three times despite both the (high) cost quoted for a further extension and new advisories from the State government barring further AMC extensions.
- In view of above, addressing any potential bottlenecks in SCADA system will not be feasible rendering the entire System operation and Commercial operation at SLDC vulnerable.
- Last week, in a meeting concerned vendor M/S GE was requested to initiate operation of ULDC-III system without any further delay. It has been assured by that at least real time display of WB SLDC and ALDC(WBSEDCL) will be completed by 30th Sept 2029.
- Concerns raised on completion of ICCP connectivity with ERLDC under new project.

The forum was apprised that a follow-up review meeting is scheduled around 7th September'26 to assess further progress on ICCP integration and state-level readiness i.r.o ULDC-III project.

OCC Decision

- OCC noted the criticality of ULDC-III readiness ahead of the 30th September ULDC-II AMC expiry and advised Powergrid to ensure the real-time display functionality at WB SLDC & ALDC is operational before that date, using a temporary/interim connectivity arrangement at WBSEDCL's ALDC premises if the permanent building/power infrastructure is not ready in time.
- All constituent states were advised to plan for parallel operation of ULDC-II and ULDC-III for an estimated three-to-five months post-commissioning, and to make necessary interim arrangements (manpower, backup systems etc).

2.16 Rehabilitation and upgradation to 4 laning of NH-31 D from Km 113+200 to Km 154+854 (Pkg-IIA) Falakata –Salsalabari Section in the state of West Bengal on EPC mode.: Request for 10-day Shutdown of High-Tension Line 220 KV D/C Alipurduar-Salakati Tower No. (186-187) for Execution of Piling Work at KM 142+980 of Viaduct Portion: NHAI

It is to bring to your kind notice that the National Highways Authority of India (NHAI) is executing the "Balance work of Rehabilitation and upgradation to 4-laning of NH-31 D from Km 113+200 to Km 154+854 (Pkg IIA) Falakata – Salsalabari Section in the state of West Bengal on EPC mode. Piling works for the Viaduct at Km 142+980 are scheduled to commence shortly. However, 19 (Nineteen) piles fall directly within the safety corridor of the PGCIL High-Tension Line 220 KV D/C Alipurduar-Salakati between Tower No. 186 and 187.

- ☐ Due to the heavy machinery involved (Piling Machine Height: ~26.1m, Crane Height: ~36.3m) and the proximity to the live HT lines (GL to Wire Height: ~26.3m to 37.2m), it is highly unsafe and impracticable to execute the piling works without a scheduled power shutdown.

- ☐ To ensure the safe execution of these critical highway infrastructure works, a daily shutdown from **06:00 Hrs to 18:00 Hrs** is urgently required for a total period of **10 Days**, tentatively starting from **26.08.2026 to 04.09.2026**. The circuit-wise shutdown plan proposed by the contractor is as follows:

Sl. No.	Pile Group	Circuit	Pile Details	Nos of pile	No of days for execution of pile	Date & Time for shutdown
1	B LHS	Circuit 1	B1 to B4	4	2 days	26.08.2026 to 27.08.2026 from 6.00 Hrs to 18.00 Hrs on daily basis
2	D LHS	Circuit 1	D5 to D8	4	2 days	28.08.2026 to 29.08.2026 from 6.00 Hrs to 18.00 Hrs on daily basis
3	A RHS	Circuit 2	A1 to A4	4	2 days	30.08.2026 to 31.08.2026 from 6.00 Hrs to 18.00 Hrs on daily basis
4	C RHS	Circuit 2	C1 to C3	3	2 days	01.09.2026 to 02.09.2026 from 6.00 Hrs to 18.00 Hrs on daily basis
5	D RHS	Circuit 2	D1 & D2	2	1 days	03.09.2026 from 6.00 Hrs to 18.00 Hrs on daily basis
6	D LHS	Circuit 1 & 2	D3 & D4	2	1 days	04.09.2026 from 6.00 Hrs to 18.00 Hrs on daily basis
Total				19	10 days	

- ☐ It is pertinent to mention that subject project is of national importance and is a part of missing link of East West Corridor of India. Further, Hon'ble Prime Minister had laid the foundation stone of subject project. The project is being monitored at highest level of NHAI/MORTH, Govt. of India and has been also included in the list of Project Monitoring Group (PMG).
- ☐ Furthermore, considering that this is an infrastructure project of paramount national importance, we kindly request you to consider waiving off the shutdown charges, if any in this special case.
- ☐ Kindly allow the shut-down as per above schedule.

Deliberations in the meeting

NHAI requested for the 220 KV D/C Alipurduar-Salakati between Tower No. 186 and 187 due to ongoing NH-31 works in the Siliguri corridor(WB). Pile foundation works are planned to be carried out in the S/D window.

OCC Decision

OCC approved the S/D as per requested schedule cum plan i.e from 26.08.2026 to 04.09.2026 for 10 days (on daily basis) subject to consent from NERPC.

3. PART-C: ITEMS FOR INFORMATION

3.1. ER Grid performance during July 2026

The average and maximum consumption of Eastern Region and Max/Min Demand (MW), Energy Export for the month July-2026 were as follows:

AVERAGE CONSUMPTION (MU)	MAXIMUM CONSUMPTION(MU)/ DATE	MAXIMUM DEMAND (MW)	MINIMUM DEMAND (MW)	SCHEDULE EXPORT	ACTUAL EXPORT
		DATE / TIME	DATE / TIME	(MU)	(MU)
667 MU	697 MU, 17.07.2026	32391 MW, 03.07.2026 at 22:58 Hrs.	23078 MW, 06.07.2026 at 06:08 Hrs.	3531	3871

Deliberations in the meeting

The grid performance of ER for the month of July 2026 was highlighted.

3.2. Non-Submission of FRC data in stipulated time-frame: ERLDC

Adhering to IEGC clauses 30.8 and 30.10.(a) to 30.10.(q), generating stations within the Eastern region are required to submit essential data to ERLDC within two days of receiving a notification regarding a reportable frequency event. Additionally, according to clause 30.10.(n), all control areas within the eastern region must assess their frequency response characteristics and share the evaluation, along with high-resolution data, with the ERLDC. Therefore, timely submission of primary response data is crucial for compliance with the IEGC.

Hence all are again requested to follow the stipulated timeline and submit the data to ERLDC and also fill the google sheet below to include the email address where notifications of reportable events should be sent.

The latest data receipt status is given below for **Jul-2026** events: (as on 10.08.2026):

STATIONS		16.07.2026 11:52 hrs	19.07.2026 12:36 hrs	25.07.2026 19:20 hrs
FSTPP #STG 1 & 2	ISGS			
FSTPP # STG 3	ISGS			
KhSTPP #STG 1	ISGS			
KhSTPP #STG 2	ISGS			
TSTPP #STG 1	ISGS			
Barh stage-1	ISGS			
Barh stage-2	ISGS			
BRBCL	ISGS			
Darlipalli	ISGS			
North Karanpura	ISGS			
NPGC	ISGS			
TEESTA V	ISGS			
PVUNL	ISGS			
Dikchu	IPP			
IBEUL (JSW UTKAL)/INDBHARAT	IPP			
GMR	CPP			
MPL	CPP			
ADHUNIK	CPP			
JITPL	CPP			
TEESTA III	CPP			
Bihar	STATE			
Jharkhand	STATE			
DVC	STATE			
OPTCL	STATE			
WB	STATE			
Updated as on	10.08.2026			
	Received			
	Not Received			
	Plant Out			
	Data freeze at plant			

Hence all are again requested to follow the stipulated timeline and submit the data to ERLDC and also fill in the google sheet below to include the email address where notifications of reportable events should be sent.

https://docs.google.com/spreadsheets/d/1slvAOmQIEQVIMn0LnB78eKMa2sz2QYICZ-sPEpeV_jk/edit?usp=sharing

- Primary Frequency Response Performance of Eastern Regional Generators and States During Events in July 2026

Event Details

Event	Date and Time	Description
Event 1	11:52 hrs. on 16.07.2026 (Thursday)	Reporting of Load loss event of 1150 MW in Andhrapradesh, SR at 11:52 hrs. on 16.07.2026
Event 2	12:36 hrs. on 19.07.2026 (Sunday)	Reporting of Generation loss event of 1150 MW in 400 kV Kanarpatty & Tirunelveli, SR at 12:36 hrs. on 19.07.2026
Event 3	19:20 hrs. on 25.07.2026 (Saturday)	Reporting of Generation loss event of 1200 MW in Greenko PSP, SR at 19:20 hrs. on 25.07.2026

Event-wise Generator Performance Summary

Generator	Event 1				Event 2				Event 3			
	Droop based on MCR %	Droop Grade	FRO %	FRO Grade	Droop based on MCR %	Droop Grade	FRO %	FRO Grade	Droop based on MCR %	Droop Grade	FRO %	FRO Grade
Adhunik	-65.6	Poor	-567.1	Poor	196.8	Excellent	1700.1	Excellent	6.1	Poor	82.8	Average
BARH ST-1	22.2	Poor	293.3	Excellent	344.1	Excellent	4541.7	Excellent	-31.7	Poor	-644.2	Poor
BARH ST-2	47.9	Poor	428.6	Excellent	579.4	Excellent	5185.1	Excellent	0.9	Poor	13.2	Poor
BRBCL	35.6	Poor	323.3	Excellent	531.4	Excellent	4831.1	Excellent	7.5	Poor	107.8	Excellent
DARLIPALLI	52.2	Below Average	407.7	Excellent	59.2	Below Average	462.1	Excellent	0.0	Poor	0.3	Poor
DIKCHU	229.0	Excellent	898.6	Excellent	156	Excellent	432.0	Excellent	25	Poor	140.5	Excellent
FSTPP I&II	-18.6	Poor	-186.4	Poor	353.4	Excellent	3534.1	Excellent	65.4	Below Average	1020.5	Excellent
FSTPP III	21.3	Poor	177.7	Excellent	655.0	Excellent	5458.3	Excellent	6.6	Poor	87.6	Good
GMR	53.9	Below Average	407.6	Excellent	518.2	Excellent	3921.8	Excellent	55.5	Below Average	647.9	Excellent
INDBHARAT	-4.0	Poor	-42.7	Poor	68.4	Below Average	736.9	Excellent	-19.3	Poor	-318.4	Poor
JITPL	-31.5	Poor	-239.7	Poor	133.9	Excellent	1020.1	Excellent	22.2	Poor	259.5	Excellent
KhSTPP I	42.0	Poor	392.3	Excellent	-260.5	Poor	-2431.2	Poor	-14.4	Poor	-209.8	Poor
KhSTPP II	44.9	Poor	369.0	Excellent	876.1	Excellent	7200.9	Excellent	3.7	Poor	47.5	Poor
MPL	68.8	Below Average	601.9	Excellent	223.4	Excellent	1954.4	Excellent	25.7	Poor	348.2	Excellent

NORTH KARANPURA	37.3	Poor	383.3	Excellent	668.3	Excellent	6874.3	Excellent	-8.6	Poor	-135.6	Poor
NPGC	14.8	Poor	130.6	Excellent	319.6	Excellent	2812.3	Excellent	-4.5	Poor	-61.4	Poor
PVUNL	0.0	Poor	0.0	Poor	178.2	Excellent	1500.8	Excellent	18.3	Poor	233.8	Excellent
TSTPS-I	54.5	Below Average	444.9	Excellent	89.6	Good	731.3	Excellent	6.7	Poor	86.5	Good
Teesta V	112	Excellent	422.1	Excellent	186	Excellent	585.8	Excellent	5	Poor	31.6	Poor

Summary:

State	Event 1		Event 2		Event 3	
	FRO Response %	FRO Grade	FRO Response %	FRO Grade	FRO Response %	FRO Grade
Bihar	332.7	Excellent	-2444.7	Poor	682.1	Excellent
DVC	481.9	Excellent	2372.5	Excellent	191.8	Excellent
Jharkhand	-657.8	Poor	741.0	Excellent	727.3	Excellent
Odisha	-163.4	Poor	-1908.8	Poor	558.9	Excellent
West Bengal	4.0	Poor	307.3	Excellent	104.2	Excellent

Poor < 50%	Below Average ≥ 50% and < 75%	Average ≥ 75% and < 85%	Good ≥ 85% and < 100%	Excellent ≥ 100%
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Deliberations in the meeting

OCC Decision: -

- All generators were advised to regularly share high resolution data against each reportable frequency event with ERLDC on time to facilitate accurate assessment of FRP for respective control areas.
- All generating utilities were also urged to update the google sheet (link mentioned above) with email address where notifications of reportable events will be shared.

3.3. Regarding Non-Submission of Forecasting and Resource Adequacy Data from States: ERLDC

- ☐ Clause 2 of Regulation 31 of IEGC 2023 has mandated all SLDCs to timely submit the demand estimate data to the respective RLDC and RPC.
- ☐ Current data submission status is given in the table below: Hence, it is again requested to all the concerned for timely submission of demand estimation data to ERLDC. This collaboration is essential for effective planning and preparedness to meet the region's electricity demands efficiently and reliably.
- ☐ Latest Forecast and Resource Adequacy Data receipt status at ERLDC is shown below:

4. PART-D: OPERATIONAL PLANNING

4.1. Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on 12-08-2026)

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	DARLIPALI	ODISHA	NTPC	1	800	Tripped on boiler tube leakage but later taken into comprehensive overhaul	03-Aug-2026
2	KHSTPP	BIHAR	NTPC	3	210	Overhauling	28-Jul-2026
3	GMR	ODISHA	GMR-KEL	1	350	Annual Overhauling.	22-Jul-2026
4	NABINAGAR(BRBCL)	BIHAR	NTPC	3	250	Annual Maintenance	13-Jul-2026
5	BARH	BIHAR	NTPC	2	660	Initially unit was out due to Boiler tube leakage, later unit was taken under Annual Overhauling from 00:00 hrs of 01.07.2026	26-Jun-2026
6	DPL	WEST BENGAL	DPL	7	300	Capital overhauling	12-Jul-2026
7	FSTPP	WEST BENGAL	NTPC	3	200	Master Fuel Trip (MFT) malfunction	11-Aug-2026
8	BARH	BIHAR	NTPC	3	660	Boiler Tube Leakage	11-Aug-2026
9	FSTPP	WEST BENGAL	NTPC	4	500	Tripped on boiler tube leakage	10-Aug-2026
10	MEJIA TPS	DVC	DVC	1	210	Boiler Tube Leakage	11-Aug-2026
11	SAGARDIGHI	WEST BENGAL	WBDCL	5	660	Taken out due to various leakage repairing jobs and Air Preheater maintenance jobs.	08-Aug-2026
12	BANDEL TPS	WEST BENGAL	WBDCL	5	215	Unit was taken out of service under extreme emergency conditions due to critical technical constraints, necessitating immediate shutdown for essential overhauling work.	27-Jul-2026
13	KOLAGHAT	WEST BENGAL	WBDCL	3	210	Due to Fire hazard in boiler	18-Jun-2026

14	RTPS	DVC	DVC	2	600	High Turbine Vibration	14-Apr-2026
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All Generating stations are requested to update expected restoration time and reason outage to ERLDC/ERPC on weekly basis in case of any change at their end.

Major Generating stations Out on Reserve Shutdown due to low system demand:

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
NA							

Hydro Unit Outage Report: -

S. NO	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	U. KOLAB	ODISHA	OHPC	1	80	Removal of old AVR and installation, testing & commissioning of new DAVR by M/s BHEL.	07-Aug-2026
2	U. KOLAB	ODISHA	OHPC	3	80	Stator Earth Fault	19-Jul-2026
3	U. KOLAB	ODISHA	OHPC	2	80	Heavy Leakage in guide vane	22-Jan-2026
4	BALIMELA HPS	ODISHA	OHPC	5	60	Renovation and modernization work	16-Jan-2025
5	BALIMELA HPS	ODISHA	OHPC	6	60	Initially unit was out due to Severe water leakage from turbine, later unit was taken under Renovation and modernisation work from 00:00 hrs of 16.01.25	06-Jan-2025
6	BURLA HPS/HIRAKUD I	ODISHA	OHPC	7	37.5	Spark observed at PMG rotor	08-Jul-2026
7	TEESTA STG III Hep	SIKKIM	TUL	1	200	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
8	TEESTA STG III Hep	SIKKIM	TUL	2	200		04-Oct-2023
9	TEESTA STG III Hep	SIKKIM	TUL	3	200		04-Oct-2023
10	TEESTA STG III Hep	SIKKIM	TUL	4	200		04-Oct-2023
11	TEESTA STG III Hep	SIKKIM	TUL	5	200		04-Oct-2023
12	TEESTA STG III Hep	SIKKIM	TUL	6	200		04-Oct-2023

4.2. Long outage report of transmission Element (MORE THAN 01 WEEK) (As on 12.08.2026):

Transmission Element / ICT	Outage Date	Reasons for Outage
220KV-FSTPP-LALMATIA-1	21-04-2021	Two nos. of tower collapsed on 29.05.2024 near to Lalmatia GSS in the Loc. No. 246 & 247. 220KV S/C Farakka-Lalmatia Transmission Line is in anti-theft charging condition from Loc no 248 (Lalmatia end) to Loc no 24. Foundation, erection and stringing in progress from loc 1 to 24.
132KV-BARHI-RAJGIR-1	25-03-2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earth wire between tension tower no. 218-237 in same line. The lines from Barhi(DVC) will be terminated at Barachatti(BH) and new line to be constructed from Barachatti to Rajgir (BH)
132KV-NALANDA-BARHI(DVC)-1	25-03-2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earth wire between tension tower no. 218-237 in same line. The lines from Barhi(DVC) will be terminated at Barachatti (BH) and new line to be constructed from Barachatti to Nalanda (BH)
132KV-CHANDIL-MANIQUEI-1	05-06-2024	Line is closed when power assistance required by either party. Presently, line is opened as power assistance not required by any party.
400KV/220KV 315 MVA ICT 1 AT NORTH KARANPURA	12-09-2024	Tripped on Differential protection.
400KV/220KV 315 MVA ICT 1 AT TSTPP	01-11-2024	Tripped on PRD protection. The failed transformer has reached the vendor; the repair scope has been finalized
400KV/220KV 315 MVA ICT 2 AT MEJIA-B	20-01-2025	315 MVA ICT-2 at MTPS-B got damaged while charging from 220kV GIS bay. New procurement of ICT has been taken up & installation of the same may complete by end of Mar'28.
400KV-DIKCHU-RANGPO-2	05-08-2025	Shutdown taken for damaged insulator replacement work. While charging the line, bus bar protection operated at Dikchu. Issue in GIS chamber of Y ph Isolator between line CB and bus 2. Powder formation inside isolator chamber has occurred. Restoration work delayed. Current Status: Line is anti-theft charged from Rangpo end to Dikhchu extended via Teesta-III, Unable to take in service due to issue in tie bay at Dikchu.

400KV MAIN BUS - 2 AT DIKCHU	05-08-2025	Bus bar protection operated while charging 400kV Dikchu-Rangpo 2. Issue in GIS chamber of Y ph Isolator between Rangpo ckt 2 line CB and bus 2. Powder formation inside isolator chamber has occurred. Restoration work delayed. Current Status: Negotiation with GE for procurement is under process, Further 6 to 8 months required by GE for order delivery.
220KV-PATNA-KHAGAU-1	02-08-2025	LBB relay operated during rectification of DC grounding defect by M/S KRR at GSS khagaul. Earlier w.e.f 02-08-2025 12:06 Hrs, Tower No. 63 has bent significantly on one side. As on 18.06.2026, Tower No. 63 has bent significantly on one side, tender has been awarded. Foundation work for tower has been completed. Expected Revival - 12/08/2026
220KV-BIDHANNAGAR-WARIA-1	29-10-2025	Line kept open to control loading of 220 kV Waria-Mejia D/C (Anti-theft charged from Waria end.)
220KV-BIDHANNAGAR-WARIA-2	29-10-2025	Initially line was opened to control line loading. In between B-phase CT Blasted at Bidhannagar end. Now Line is charged as anti-theft from Waria end to control loading of 220 kV Waria-Mejia D/C.
220KV-BALIMELA-UPPER SILERU-1	21-11-2025	Idle charged from U. Sileru end. Power drawl by Odisha halted due to non-concurrence by Andhra Pradesh.
132KV-MADHEPURA (BH)-SAHARSA-1	18-12-2025	To control the line loading. Line kept idle charged from Saharsa.
400KV/220KV 315 MVA ICT 1 AT KODERMA	10-02-2026	The ICT had been taken out of service due to a rising trend of acetylene (C ₂ H ₂) concentration in the transformer oil. The ICT was proposed to be sent to the factory for necessary rectification. Meanwhile, one ICT from DSTPS was diverted and installed at Koderma. After commissioning, the diverted ICT was idle-charged from the 400 kV side at 17:09 Hrs on 02.08.2026. Subsequently, Dissolved Gas Analysis (DGA) of the transformer oil sample indicated a rising trend of acetylene (C ₂ H ₂) concentration. Accordingly, the ICT was taken under shutdown at 18:11 Hrs on 04.08.2026 for carrying out further investigation and necessary corrective action.
132KV-RANGIT-SAGBARI-1	28-02-2026	Line was taken out to carry out necessary jumpering modification at Sagbari station to connect directly to Legship 220/132/66kV. The FTC of the line is pending in absence of necessary statutory clearance.

400KV-TEESTA-III-RANGPO-1	15-03-2026	Line is anti-theft charged from Rangpo end to Dikhchu extended via Teesta-III
400KV/220KV 315 MVA ICT 2 AT DSTPS(ANDAL)	03-05-2026	ICT has been shifted to KTPS.
132KV-PATRATU-PATRATU-2	09-05-2026	For system requirement power order change
400/220 KV 315 MVA ICT 4 AT ARAMBAGH	29-05-2026	Tripped on buchholz protection.
220KV-KATAPALLI-BOLANGIR(PG)-1	17-06-2026	For system requirement
220KV-Rongnichu-RANGPO-1	18-06-2026	Dead end Tower No. 30 (at the boundary of Rangpo s/s) from the Rongnichu end has collapsed
765KV-JHARSUGUDA-RAIPUR PS (DURG)-2	05-07-2026	H/T ON VOLTAGE REGULATION
220KV-SAHARSA(PMTL)-KHAGARIA(NEW)-1	09-07-2026	Khagaria: Z-1, Y_E fault, FD-65 KM, FC-2.6 KA. During tower-top patrolling, multiple polymer insulators found punctured.
220KV-SAHARSA(PMTL)-KHAGARIA(NEW)-2	14-07-2026	Distance protection operated Bihar: R phase, zone 1 , dist-52.77km IR-1.798kA IY-30.61A IB-27.60A. During tower-top patrolling, multiple polymer insulators found punctured.
765KV-DHARAMJAIGARH-JHARSUGUDA-2	21-07-2026	H/T ON VOLTAGE REGULATION
400KV-BARH-MOTIHARI-2	21-07-2026	For tower dismantling and erection of tower at pile foundation at Location No. 27 (Athmal Gola, Barh, Patna).

400KV-BARH-MOTIHARI-1	21-07-2026	For tower dismantling and erection of tower at pile foundation at Location No. 27 (Athmal Gola, Barh, Patna).
765KV-ANGUL-JHARSUGUDA-4	22-07-2026	H/T ON VOLTAGE REGULATION
132KV-NAGARUNTARI-NABINAGAR-1	28-07-2026	For re-sagging of conductor.
220KV-JAYNAGAR-JEYPORE-3	29-07-2026	Line tripped at 03:44 Hrs. After charging, R phase current was very less at both jayanagar and PGCIL end. Line handtripped.
765KV-NEW RANCHI-DHARAMJAIGARH-2	02-08-2026	Line diversion work between tower no. 807 to 811 and 663 to 666 due to construction of NH-130A (URGA-PATHALGAON SECTION)

Deliberations in the meeting

Members noted.

4.3. Commissioning of new units and transmission elements in Eastern Grid in the month of July-2026

The details of new units/transmission elements commissioned in the month of July-2026 based on the inputs received from beneficiaries:

NEW ELEMENTS COMMISSIONED DURING July, 2026							
उत्पादन इकाइयाँ / GENERATING UNITS							
क्र. सं. Sl. No.	स्थान Location / Pooling Station	मालिक/यूनिट का नाम OWNER/UNIT NAME	यूनिट संख्या/स्रोत Unit No/Source	संकलित क्षमता (मेगावाट) Capacity added (MW)	कुल/स्थापित क्षमता (मेगावाट) Total/Installed Capacity (MW)	दिनांक DATE	टिप्पणी Remarks
1	TEESTA HPS	TEESTA HPS - UNIT 1	Hydro	170	510	12-07-2026 18:38	These units have been restored and recommissioned following the Glacial Lake Outburst Flood (GLOF) that occurred in October 2023.
2		TEESTA HPS - UNIT 2	Hydro	170	510	12-07-2026 21:02	
3		TEESTA HPS - UNIT 3	Hydro	170	510	15-07-2026 17:00	
आई.सी.टी./जी.टी./एस.टी. / ICTs/ GTs / STs							

क्र . सं. Sl. No .	एजेंसी/ मालिक Agency/ Owner	उप-केन्द्र SUB-STATION	आईसीटी संख्या ICT NO	वोल्टेज (केवी) Voltage Level (kV)	क्षमता (एमवीए) CAPACITY (MVA)	दिनांक DATE	टिप्पणी Remarks
1	NIL						
प्रेषण लाइन / TRANSMISSION LINES							
क्र . सं. Sl. No .	एजेंसी/ मालिक Agency/ Owner	लाइन का नाम LINE NAME	लंबाई (किमी) Length (KM)	कंडक्टर प्रकार Conductor Type	दिनांक DATE	टिप्पणी Remarks	
NIL							
लिलो / प्रेषण लाइन की पुनर्व्यवस्था / LILO/RE-ARRANGEMENT OF TRANSMISSION LINES							
क्र . सं. Sl. No .	एजेंसी/ मालिक Agency/ Owner	लाइन का नाम / लिलो पर Line Name/LILO at	लंबाई (किमी) Length (KM)	कंडक्टर प्रकार Conductor Type	दिनांक DATE	टिप्पणी Remarks	
1	PGCIL	400 kV New Purnea–Farakka S/C	171.62	ACSR Triple Snowbird	29-07-2026 12:05	The above transmission lines were charged as the 400 kV Farakka–Gokarna S/C line under an interim bypass arrangement to facilitate the execution of transmission strengthening works by POWERGRID.	
2	PGCIL	400 kV New Purnea–Gokarna S/C	248.71	ACSR QUAD MOOSE	29-07-2026 13:19		
3	PGCIL	400KV-GAYA-KODERMA-1	125.966	ACSR QUAD MOOSE	27-07-2026 00:02	These D/C transmission lines were temporarily charged as the 400 kV Koderma–Maithon D/C transmission line for carrying out transmission strengthening works by PGCIL.	
4	PGCIL	400KV-GAYA-KODERMA-2	125.966	ACSR QUAD MOOSE	27-07-2026 00:05		
5	PGCIL	400KV GAYA-MAITHON-1	275.665	ACSR QUAD MOOSE	27-07-2026 00:20		
6	PGCIL	400KV GAYA-MAITHON-2	01-10-1900	ACSR QUAD MOOSE	27-07-2026 00:21		
7	BSPTCL	132KV-SAHARSA(PMTL)-Saharsa-1B(BH)-1	30	Single Panther	24-07-2026	LILO of Saharsa-1B and Banmankhi(BH) at 132 kV Saharsa(PG)	
बस/लाइन रिएक्टर / BUS/LINE REACTOR							

क्र . सं. Sl. No .	एजेंसी/ मालिक Agency/ Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	रेटिंग (एमवीएआर) Rating (MVAr)	दिनांक DATE	टिप्पणी Remarks
1	NIL					
एच.वी.डी.सी/ए.सी फिल्टर बैंक/फैक्ट्स डिवाइस संबद्ध प्रणाली / HVDC /AC Filter bank / FACTS DEVICE associated System						
क्र . सं. Sl. No .	एजेंसी/ मालिक Agency/ Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks
NIL						
बस - बे / BUS - BAYS						
क्र . सं . S l.N o.	एजेंसी/ मालिक Agency/ Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks
1	NHPC	400 kV GIS Bay: 405 (Bus Coupler Bay) at Teesta V	TEESTA V	4006V	29-07-2026	
2	NHPC	400 kV GIS Bay: 404 (Rangpo-II Bay) at Teesta V	TEESTA V	4006V	10-07-2026 01:15	
3	NHPC	400 kV GIS Bay: 406 (Rangpo-I Bay) at Teesta V	TEESTA V	4006V	29-07-2026	
NIL						

Deliberations in the meeting

Members noted.

4.4. UFR operation during the month of July 2026

Frequency profile for the month as follows:

MONTH	MAX (DATE/TIME)	MIN (DATE/TIME)	% LESS IEGC BAND	% WITHIN IEGC BAND	% MORE IEGC BAND
July 2026	50.35 (on 01-July-26 at 13:06 Hrs.)	49.42 (on 25-July-26 at 19:20 Hrs.)	8.8	72.5	18.7

Hence, no report of operation of UFR has been received from any of the constituents.

Deliberations in the meeting

Members noted.

Meeting ended with a vote of thanks to the chair.

Annex-A

Participants in 242nd OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 20.08.2026 (Thursday)

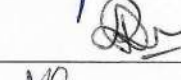
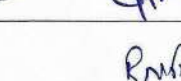
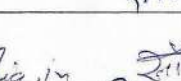
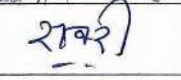
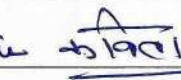
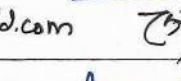
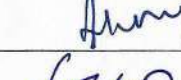
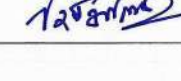
Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
1	K. B. Jagtap	Member Secretary	ERPC	8652776033	mserpc-power@nic.in	10/08/2026
2	S Banerjee	Executive Director	ERLDC	9433041823	surajit.banerjee@grid-india.in	20/08/2026
3	R.K Meena	Director	ERPC	8800764840	rkmeena@erpc.in	20/08/2026
4	S C De	CGM	ERLDC	9436335369	s.c.de@grid-india.in	20/08/2026
5	D. BISWAS	GM	ERLDC	9434740041	dbiswas@grid-india.in	20/08/2026
6	Bikash Achari	DGM	ERLDC GRID-INDIA	7003472016	bikash.achari@grid-india.in	20/08/2026
7	Mahavir Prasad Singh	GM	ERLDC	9729872327	mahavir@grid-india.in	20/08/2026
8	Manas Das	DGM	ERLDC GRID-INDIA	9007070925	manasdas@grid-india.in	20/08/2026
9	Harekrishna Jena	CGM	NTPC-Barh	9437048589	hkjena@ntpc.co.in	20/08/2026
10	C.B. OJHA	GM	NTPC-ERLDC PATNA	9650950709	CHANDRA BHUSHAN OJHA@NTPC.CO.IN	20/08/2026
11	SHYAMAL MONDAL	AGM	NTPC-Kahalgaon	9406712572	shyamalmondal@ntpc.co.in	20/08/2026
12	AKSHAY KUMAR SIVASTAVA	AGM	NTPC-BARH	9650908719	akshaykumar@ntpc.co.in	20/08/2026
13	Arshad Jilani	DEM	NTPC Kolkata	9650997632	arshadjilani@ntpc.co.in	20/08/2026
14	SHOUVIK BANERJEE	CE, SLDC	SLD/WBSETCL	9434910379	ce.wbsetcl@gmail.com	20/08/2026
15	Preetam Banerji	ACE	WBSEDCL	7003871189	preetam72@gmail.com	20/08/2026

Participants in 242nd OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 20.08.2026 (Thursday)

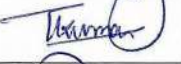
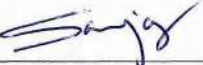
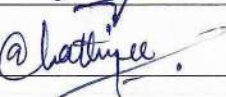
Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
16	J. Sengupta	ACE, CPD	WISSETEL	9434910547	cpd.wbsetel@gmail.com	
17	M. Mondal	AG, M(OS)	WBPDCL	8336903808	m.mondal@wbpdcl.co.in	
18	Rajat Kr. Koley	SM(OS)	WBPDCL	9474860642	rk.koley@wbpdcl.co.in	
19	Manas Guha	DGM(E)	DPL	8509267341	m.guha@dpl.net.in	
20	SURAJIT ROY	Chief Mgr	HEL	8335067270	surajit.roy@rpsg.in	
21	DEBASWEE DE	DGM	CESC	92305-21123	debasw.de@rpsg.in	
22	ARUN BANDYOPADHYAY	SR. Manager.	CESC	9163312743	arun.bandyopadhyay@rpsg.in	
23	Goutam Verma	D.M	JIPL	7565832211	JIPL Scheduling @ Jindal group.co	
24	Rajnarayan Tiwari	Sr. manager	JIPL	7987898248	" — "	
25	Sourav Mandal	CM	ERLDC	9402102354	souravmandal@grid-india.in	
26	Shabari Pramanick	CM	ERLDC	9007058764	shabari.pramanick@grid-india.in	
27	KAVITA PARIHAR	G. M.	ERLDC	9958114989	kavita@grid-india.in	
28	Somnath Mukherjee	A. M.	Indignid.	7001389952	somnath.mukherjee@indignid.com	
29	Anupriya Choudhury	Gr. Mgr	Jiw Indus ment	9937294336	Anupriya.choudhury@jiw.in	
30	CHANDAN MALICK	C.M (SO)	ERLDC, GRID- INDIA	7278756275	chandan.mallick@grid-india.in	

Participants in 242nd OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 20.08.2026 (Thursday)

Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
31	ALOK PRATAP SINGH	C. M.	GRID-INDIA	887470777	apksingh@grid-india.in	
32	GAGAN KUMAR	E. Ex. E.	SLDC, BSPTCL	7992486100	gagankumishra@gmail.com	
33	TUNESHWAR KUMAR	SR. MANAGER	STU, JUSNL	9431190306	tuneshwar.kumar@gmail.com	
34	Rinil Topno	SR. Manager	SLDC, JUSNL	9835715518	slcdcanchi@gmail.com	
35	Alok Kumar Moharana	GM (EI)	OPTCL, BBSR	9438907246	ele.akmaharana@optcl.co.in	
36	Sanjaya Kumar Mishra	CGM (PP)	BIRJ DCO	9438907414	sanjayasld@gmail.com	
37	Manas Ranjan Samantary	Sr. Asst. (GD)	SLDC.	9438907713	manasranjan789@gmail.com	
38	Bikash Ranjan Bera	Asst. M	SLDC	9438907212	bikash986@gmail.com	
39	Sanjay Kumar Sharma	HoD, Sr GM	DVC LDC.		sanjay.sharma@dvc.gov.in	
40	S. K. Panda	manager	DVC LDC	6370134794	santosh.panda@dvc.gov.in	
41	AMIT SARUAR	Manager (J)	NHAI	9123861434	soholnhai@gmail.com	
42	PARTHA GHOSH	DM (AM)	ERPD	9434748263	partha.ghosh@poweroid.in	
43	RAKESH KR. PRADHAN	DGM	GRID-INDIA	9831337570	rkpradhan@grid-india.in	
44	Narender Kumar	M. D.	NHA D/O CC.	9728104848	narender.singh@nhai.co.in	
45	Agniva Chatterjee	AD-I	ERPC			

Annex B.2.1.(a)

Logic of bus voltage selection etc seems in order. Following are the observations regarding logic:

Item	Proposed scheme	Attached PDF	Observation and comments
Use minVolt = Min(Bus1_VLL_AVG, Bus2_VLL_AVG)			Minimum voltage across the six/three line voltages should be taken after validation of voltage, not the minimum of avg. voltages of the two bus.
Stage-1 threshold	<370 kV	<370 kV	OK
Stage-1 persistence	5 s	25 s	Too large by that time under dynamic conditions collapse may occur.
Stage-1 load shedding	Stage-1 load-200	Group 32 ~200 MW	OK
Recovery after Stage-1	≥375 kV → stop	≥375 kV → reset	OK
Mandatory stabilization after Stage-1	No 40-s wait proposed	40 s	The proposed stabilization delay is excessively large and is not recommended from a voltage-stability perspective. Under dynamic conditions, such a prolonged delay without any further corrective action may allow the voltage to deteriorate further, potentially leading to voltage collapse. Therefore, the stabilization delay should preferably be eliminated altogether. If a stabilization period is considered essential, it should be restricted to not more than 3–4 seconds.
Stabilization Block			Stabilization Block may either be completely removed or if kept then it should be placed after operation of Stage-1 in the process flow diagram.
Stage-2 criterion	Still <375 kV after Stage-1	<375 kV	OK
Stage-2 persistence	2 s	15 s	This delay is also excessively large. If Stage-2 operation is required, it indicates that the system conditions have not improved sufficiently even after Stage-1 load shedding; therefore, further corrective action should not be unnecessarily delayed. A 2-second delay for Stage-2 operation is considered adequate. As per the proposed logic in the document, Stage-2 may operate only after 25 seconds of Stage-1 delay + 40 seconds of stabilization wait + 15 seconds of Stage-2 delay, i.e., approximately 80 seconds from the initial voltage depression. Such a

			prolonged delay is detrimental to system stability and may aggravate the risk of voltage collapse. Exact starting point of Stage-2 timer needs to be specified. It should mention second Stage-2 timer begins from: • issuance of the Stage-1 trip command,
Stage-2 load	200 MW	Group 31; MW not clearly stated	Should explicitly specify approximate quantum MW

Annex B.2.1.(b)

Status of DVC Units under Phase-I Implementation

As per the phasing plan notified under the CEA (Flexible Operation of Coal based Thermal Generating Units) Regulations, 2023, BTPS Unit #A (500 MW) and RTPS Unit #2 (600 MW) were covered under Phase-I, with the implementation period from July 2024 to June 2026.

1. Flexible Operation Solutions at BTPS Unit #A (500 MW)

- To enable flexible operation and compliance with CEA requirements, Work Order No. GEMC-511687795492297 dated 19.12.2024 was awarded to M/s BHEL for implementation of flexible operation solutions at BTPS Unit #A.

The following major works have been completed:

- APC system commissioning activities were carried out by BHEL PSER and BHEL-EDN engineers during June 2026.
- **Stable operation at 40% TML without oil support through APC was successfully demonstrated on 15.06.2026 for 1 hour 30 minutes and on 17.06.2026 for 3 hours, with all process parameters remaining within acceptable limits during flexibility trial test in coordination with M/s BHEL.**
- **Ramp rate trials were completed with achieved ramp rates of:**
 - **2.93% in 70% to 100% TMCR range**
 - **2% in 55% to 70% TMCR range**
 - **1% in 40% to 55% TMCR range**
- Stall logic for FD Fans and PA Fans has been incorporated in DCS, along with alarm configuration and display on respective DCS pages.
- In one TDBFP, the existing ON/OFF recirculation valve has been replaced with a modulating control valve, and associated control logic has been implemented.
- Materials for BOSMON system have been received at site. Installation and commissioning are planned during the next scheduled overhauling of the unit from 25.03.2027 to 23.04.2027.

2. Flexible Operation Solutions at RTPS Unit #2 (600 MW)

RTPS Unit #2 is the first 600 MW SEC-manufactured unit in DVC. The Boiler, Turbine and Generator were supplied by Shanghai Electric Corporation (SEC), China. hence, before any upgradation/retrofitting, feasibility test for flexible operation has been

considered before implementing any retrofit or upgradation measures for flexible operation.

Accordingly, Work Order No. GEMC-511687782454419 dated 10.11.2025 was awarded to M/s Siemens Energy India Limited for carrying out a Feasibility Test for Flexible Operation of RTPS Unit #2.

The feasibility test is scheduled to be conducted in Sept 2026, after completion of the ongoing Capital Overhauling of the unit.

Further, in the 56th TCC Meeting held on 25.05.2026, DVC informed ERPC that RTPS Unit #2 is facing challenges due to non-availability of OEM support and tuning for flexible operation has not yet been undertaken. DVC also informed that MTPS Unit #7 has successfully demonstrated 40% MTL operation for 4 hours and requested that MTPS Unit #7 may be considered as a substitute for RTPS Unit #2 for meeting CEA flexible operation requirements.

In the Deliberation, TCC advised DVC to expedite implementation of 40% MTL operation at the earliest and agreed that MTPS Unit #7 may be considered as a substitute for RTPS Unit #2 to meet CEA flexible operation requirements, while ensuring continuous operation of units as per system requirement, particularly during non-solar hours.

3. Flexible Operation Solutions at MTPS Unit #7 (500 MW):

Mejia TPS Unit #7 (500 MW) has successfully demonstrated operation at 40% Technical Minimum Load (TML) and sustained the load continuously for four hours, in line with CEA guidelines for flexible operation.

Present Status:

- APC System installation done. Installation of APC panels has been completed, and control loop configuration and tuning activities carried out.
- **On 26.09.2025, the unit successfully achieved a technical minimum load of 200 MW (40% TML) without oil support and sustained operation continuously for (4) four hours, in compliance with CEA flexible operation guidelines during flexibility trial test in coordination with M/s BHEL.**
- Stall Protection installed in MTPS U#7, Commissioning is to be done.

- In both TDBFP, the existing ON/OFF recirculation valve has been successfully replaced with a modulating control valve, and the associated control logic has been implemented.
- Materials for the BOSMON system have been received and installed at site. Thermocouples have been installed in Boiler during overhauling. Commissioning is to be done.

Issues noted during Flexible operation and running at 40% TML of DVC Units

1. Coal Quality Issues:

DVC units receive coal from multiple sources with significant variations in Gross Calorific Value (GCV), Volatile Matter (VM), ash content, and moisture. These variations adversely affect combustion stability, flame sustainability, and boiler performance at low loads, making it difficult to reliably maintain unit operation at 40% TML on a continuous basis.

2. Although the recommendations given by the OEM, Bharat Heavy Electricals Limited, for flexible operation have been implemented, several operational challenges are noted during operation at 40% Technical Minimum Load (TML) and during ramping operations, as detailed below:
 - During ramp-up and ramp-down operations (2–3% loading/unloading), the Superheater (SH) temperature spray control system continues to face difficulty in maintaining Main Steam (MS) and Reheater (RH) temperatures within the permissible standard deviation limits. This ultimately imposes additional thermal stress on boiler pressure parts, turbine components, and other critical equipment. At 200 MW, the HRH steam temperature reduced to approximately 499–500°C, which was below the design operating requirement.
 - Flame stability issues are frequently encountered during low-load operations with low VM Coal.
 - Consecutive ramp-up and ramp-down operations in successive 15-minute blocks are generally not feasible. The unit requires adequate stabilization time before initiation of the next ramping cycle; hence, a suitable stabilization period is essential between successive ramping operations.
 - During ramp-up and ramp-down operations, Boiler metal temperature excursions have been observed in the Divisional Panel Superheater tubes, which may adversely affect the residual life and long-term reliability of these components.
 - IP Control valve throttling: Below 250 MW, IPCV throttling commenced. At 200 MW, both IPCVs were reported throttled to approximately 65% in MTPS U#7.

- Proper distribution of coal flow among the running feeders could not be maintained. Higher loading of the base mill was required for flame stability.

The above operational constraints associated with frequent ramp-up and ramp-down operations increase the likelihood of Boiler Tube Leakage (BTL) incidents and impose additional thermal and mechanical stresses on critical boiler and turbine components, thereby adversely affecting unit performance, reliability, operational safety, and long-term equipment health.

3. Continuous operation at 40% TML with results in increased thermal and mechanical stress on boiler and turbine components. Such operating conditions accelerate component degradation and fatigue, thereby affecting equipment reliability and potentially reducing the overall service life of the plant.

Annex B.2.1.f ଓଡ଼ିଶା ବିଦ୍ୟୁତ୍ ଶକ୍ତି ସଂଚାରଣ ନିଗମ ଲି.8.**ODISHA POWER TRANSMISSION CORPORATION LIMITED**

(A Government of Odisha Undertaking)

O/O THE CHIEF GENERAL MANAGER (O&M), BHUBANESWAR**Registered Office: Janpath, Bhubaneswar- 751007**Telephone: (0674) 2542479 (EPABX), Website: www.optcl.co.inCIN - U40102OR2004SGC007553, e-mail: cgm.onm@optcl.co.in**TW-CGM (O & M)-168/2026/1560⁽³⁾****Dt. 21/08/2026****To**

The Member Secretary
 Eastern Regional Power Committee (ERPC)
 14, Golf Club Road, Tollygunge, Kolkata - 700033

Sub: Submission of status of EHT lines undertaken for HTLS reconductoring by OPTCL - Compliance to deliberation in 242nd OCC Meeting.

Ref: 242nd OCC Meeting of ERPC held on 20.08.2026 at ERPC, Kolkata.

Sir,

With reference to the deliberation in the **242nd OCC Meeting of ERPC held on 20.08.2026 at Kolkata**, the status of EHT transmission lines undertaken by OPTCL for reconductoring /upgradation with HTLS conductor was apprised during the meeting.

As desired by the Chair, the **consolidated status of the EHT lines taken up for HTLS reconductoring / upgradation under OPTCL network**, indicating the present stage of tendering/award/execution and the scheduled/target completion, has been compiled and is enclosed herewith for kind information of ERPC.

The same may kindly be taken on record as **compliance to the deliberation of the 242nd OCC Meeting**.

Yours faithfully,

Chief General Manager (O&M)**Encl.:** As above.**Copy to:**

1. Sr.PS to CMD, OPTCL for kind appraisal of CMD.
2. Director (Operation), OPTCL, Bhubaneswar - for kind information.

HTLS RECONDUCTORING STATUS_OPTCL		
Sl. No.	Project	Status
1	400kV IB-Lapanga HTLS Up-gradation Estimated Cost: ₹ 97.82 Cr.	Re-tender floated on 13.04.2026 TC opened on 23.06.2026 Schedule for Completion: 10Months from award of LOA
2	220kV Lapanga-Katapali HTLS Up-gradation (47.52RkM) Estimated Cost: ₹ 82.08 Cr.	Price Bid Opened on 10.08.2026 Schedule for Completion: 10Months from award of LOA
3	220kV Lapanga-Budhipadar HTLS Up-gradation (18.2 RKM) Estimated Cost: ₹ 32.75 Cr.	Price Bid Opened on 10.08.2026 Schedule for Completion: 06Months from award of LOA
4	HTLS Upgradation in 132kV Burla-Katapali (Ckt-I & II) (OTSSP-II Scheme)	LOA Placed on 05.08.2026 to M/s Laser Power Schedule for Completion: 06 Months from LOA
5	Stringing of HTLS Conductor(9.12 rkm) after dismantling of ACSR Panther Conductor in existing 132 KV Lapanga-Aryan Viraj-Shyam DRI SC line along with up-gradation of requisite bay equipment in Jharsuguda District on Turnkey contract basis	LOA Placed on 14.07.02026 to M/s Paramount Schedule for Completion: 06 Months from LOA
6	Stringing of HTLS Conductor after dismantling of twin AAAC Moose Conductor in existing 400 KV Meramundali-Mendhasal DC Line along with up-gradation of equipment and Bus at respective Grid Sub-station ends (Approx. Line Length- 100.5 Route KM) on Turnkey contract basis	TC opening extended to 23.08.2026.
7	132kV Katapali-Thuapali-Bargarh (Awarded to: M/s Paramount Communications Ltd.) Awarded Cost: ₹ 43.76 Cr.)	W.O. Awarded to: M/s Paramount Communications Ltd. on 15.04.2026 Preliminary Survey Completed.
8	132kV Cuttack - Jagatsinghpur line HTLS Up-gradation EPC: M/s Sterlite Power Tr. Ltd., LoA - 31-12-2024, SoC - 30-06-2025 Cost:- ₹23.52 Cr.	Work Under Progress Foundation : 3/3 completed, Erection: 3/3 Stringing: 26.574/35.65 CKM All materials available Target: 31.07.2026
9	132kV Angul-Chainpal line HTLS Up-gradation (14.9KM) EPC: M/s Sterlite Power Tr. Ltd.,	Materials available. 132kV Angul-Chainpal line: Stringing: 12.5/14.9kM


 Chief General Manager (O&M)
 OPTCL, Bhubaneswar

Annex B.2.1.g

STANDARD OPERATING PROCEDURE ON DIVERSION of REGIONAL SPARES (TRANSFORMERS and REACTORS)

August 2026

NATIONAL POWER COMMITTEE

1. Background & Objective

POWERGRID procures and maintains regional spares transformers and reactors- to ensure the continuity of power supply and grid stability in the event of a contingency, such as the sudden failure of an in-service transformer or reactor. The primary purpose of these spares is to meet contingencies/emergencies within ISTS substations of the region. Regional spares are primarily sanctioned for use in Inter-State Transmission System (ISTS) substations. However, diversion of such spares to a State Transmission Utility (STU) may be permitted only under exceptional circumstances, based on the criticality of the requirement and subject to expeditious replenishment by the Borrower.

These Guidelines sets out the framework, eligibility, processes, roles, responsibilities, and conditions under which regional spares- transformers and reactors-maintained by POWERGRID may be diverted to eligible utilities. These Guidelines seeks to balance the need to extend emergency support to constituents with the responsibility of ensuring adequate spare availability for the ISTS network.

2. Scope & Applicability

2.1. These Guidelines applies exclusively to transformers and reactors approved by the respective Regional Power Committee (RPC) as *Regional Spares* for use in Inter-State Transmission System (ISTS) substations within the region. These spares are procured, owned, and maintained by POWERGRID to ensure operational continuity of the ISTS network during contingencies. Diversion may be considered only in case of failure of existing in-use equipment and where diversion is essential for grid security and reliability. Diversion for commissioning of new assets or Inter-regional diversion of equipment to any constituent **may also be considered provided that such commissioning is necessary to address system security, operational reliability, or other critical grid requirements. The concerned RPC shall evaluate each request on the basis of technical justification, system requirements, and the merits of the case.** All diversions may be executed strictly on a replenishment basis and may not amount to sale or transfer of ownership of the equipment.

2.2. Eligible Beneficiaries: Diversion of Regional Spares may be considered for the following **eligible beneficiaries** (hereinafter referred as Borrower) within the same Region, subject to the provisions of this SOP:

- i. ISTS Substations excluding Private Transmission Service Providers (TSPs)**
- ii. State Transmission Utility (STU) Substations.**
- iii. Private Transmission Service Providers (TSPs) operating Inter-State Transmission System (ISTS) assets as decided by concerned RPC on case-to-case basis.**
- iv. Any other utility specifically approved by the concerned RPC under exceptional circumstances.**

2.3. General Conditions Applicable to All Eligible Beneficiaries: Diversion of Regional Spare Transformers and Reactors to any eligible beneficiary shall be considered only under exceptional circumstances and shall be subject to the following conditions:

- i. The failed equipment forms part of an operational ISTS asset;
- ii. The contingency materially affects grid security, reliability, power evacuation, transfer capability or continuity of transmission services;
- iii. The concerned TSP demonstrates that no suitable spare equipment is available within its own inventory;
- iv. The diversion is approved by the concerned RPC after considering the merits of the case.

The diversion of Regional Spares serves as a facilitative measure to support system security, operational reliability, continuity of power supply, and other critical grid requirements. The concerned RPC shall consider each request on its merits, with due regard to the availability of Regional Spares, technical justification, and the larger interest of ensuring reliable, secure, and resilient operation of the regional grid.

3. Ownership and Custody of Equipment

3.1. Ownership: Ownership of all Regional Spare Transformers and Reactors shall remain vested with POWERGRID at all times. Diversion of equipment shall not:

- i. Constitute sale of equipment;
- ii. Transfer ownership rights;
- iii. Create any leasehold, mortgage, charge, lien, encumbrance, or proprietary interest in favor of the Borrower.

3.2. Custody of Equipment: The Borrower shall act only as custodian of the diverted equipment during the diversion period. The Borrower shall not:

- i. Transfer the equipment to any third party;
- ii. Create any security interest on the equipment;
- iii. Dispose of the equipment;
- iv. Modify equipment specifications without prior approval of POWERGRID.

4. Roles and Responsibilities

4.1. Lender (POWERGRID): POWERGRID may ensure that the regional spares- Transformers and Reactors- are maintained in operationally ready condition and available for deployment during contingencies. POWERGRID may:

- i. Maintain a central inventory of all Regional Spares with technical particulars and locations.
- ii. Implement the diversion only after RPC or its sub-forum's approval, as the case may be, and in accordance with this SOP.

- iii. Record every approved diversion in the central register, including date of diversion, borrower, and approved return timeline.
- iv. Monitor adherence to approved diversion timelines and report deviations to the respective RPC Secretariat.

4.2. Borrower: The Borrower may be responsible for safe custody and operation of the diverted Regional Spare during its possession. The Borrower may:

- i. Submit a formal diversion request to the respective RPC Secretariat with required technical and contingency details.
- ii. Jointly verify the equipment condition at POWERGRID substation prior to MoU signing.
- iii. Bear all expenses associated with transportation (both ways), transit insurance, erection, testing, commissioning, and related statutory charges, as applicable, and any incidental expenditure or loss to POWERGRID.
- iv. Ensure site readiness and compatibility of the spare equipment before diversion.
- v. Furnish a valid Bank Guarantee equal to the prevailing cost of the equipment, effective till 45 days after return or replenishment is completed.
- vi. Maintain and operate the equipment in accordance with applicable technical standards and POWERGRID's guidelines.
- vii. Return or replenish the equipment in healthy condition within the approved timeframe and bear any repair or replacement cost arising from damage or failure.
- viii. After returning of equipment, all pre-commissioning tests may be jointly performed at POWERGRID station to ascertain healthiness. In case of any deviation, POWERGRID may take up the repair of equipment and cost of the repair may be borne by the Borrower.

4.3. RPC Secretariat: The concerned RPC Secretariat may facilitate deliberation of the diversion proposal in the RPC or its sub-forum and place the Borrower's request before the members for decision. It may record all decisions, including timelines for return or replenishment. RPC Secretariat may communicate the decision of the forum to borrower and Powergrid within **at the earliest of decision** in RPC or its sub-forum meeting.

4.4. Reporting Requirement: The Borrower shall submit periodic reports regarding the diverted equipment. The reporting framework shall include:

- i. Monthly operational status report;
- ii. Health condition report;
- iii. Details of outages, defects, failures, repairs, or maintenance undertaken;
- iv. Progress of procurement and replenishment activities;
- v. Expected return schedule.

POWERGRID may place the status of all diverted equipment before RPC or its sub-forum at **monthly** intervals. Failure to submit reports may be treated as non-compliance under the SOP.

5. Procedure for Diversion

5.1. Submission of Request: The Borrower may submit a written request to the concerned RPC Secretariat, specifying:

- i. The nature of contingency or failure necessitating diversion, including details of affected substation and equipment.
- ii. Technical parameters of the failed equipment and the matching requirement from the Regional Spare pool.
- iii. A detailed action plan and proposed timeline for return or replenishment of the diverted equipment.

5.2. Evaluation and Approval: The RPC or its sub-forum may evaluate the request on merit, considering:

- i. The severity and genuineness of the contingency.
- ii. Availability of the Regional Spare and its necessity for ISTS grid reliability.
- iii. Impact on overall system stability and adequacy of remaining spares.

The decision of the RPC or its sub-forum will be final and binding on all concerned entities. Upon approval, the forum may record the diversion details and the agreed period of utilization in its proceedings.

5.3. Utilization Restriction: The diverted equipment shall be utilized strictly for the purpose, substation, and contingency approved by RPC. The Borrower shall not:

- i. Shift the equipment to another location;
- ii. Use the equipment for any project not approved by RPC or its sub-forum;
- iii. Further lend, lease, transfer, or assign the equipment to any third party;
- iv. Utilize the equipment for commissioning of new assets unless specifically approved by RPC.
- v. Any deviation shall require prior approval of RPC or its sub-forum.
- vi. Violation of this clause may result in immediate recall of the equipment and invocation of applicable penalties.

5.4. Execution of Memorandum of Understanding (MoU)

- i. Following the approval of diversion by the respective RPC or its sub-forum, the Borrower and POWERGRID may execute a Memorandum of Understanding
- ii. (MoU) within **thirty (30)** days from the date of RPC or its sub-forum. *For the purpose of these Guidelines, date of communication by RPC secretariat of the decisions of the forum to Borrower and Powergrid as defined in para 4.3 shall be treated as the date of RPC or its sub-forum approval, as the case may be.*
- iii. The MoU may specify all terms and conditions of diversion, including equipment details, responsibilities of both parties, financial implications, Bank Guarantee (BG) requirements,

maintenance and reporting obligations, timelines for return/replenishment, and applicable penalties for default.

5.5. Condition Assessment and Testing

- i. **Inspection before Diversion:** Before diversion, a joint inspection of the equipment **shall** be carried out by POWERGRID and the Borrower at the designated POWERGRID substation. The physical condition and test results **shall** be documented and signed jointly.
- ii. **Inspection after Return:** Upon return, the equipment shall undergo testing at the Borrower /POWERGRID station to ascertain its healthiness. Any deviation, defect, or damage observed during inspection **shall** be rectified at the Borrower's cost. Where repair or refurbishment is necessary, the Borrower **shall** carry it out through the OEM or other approved agency as per POWERGRID specifications.
- iii. **Replacement Equipment Specification:** Where replenishment is required, the replacement equipment shall:
 - a) Be of equivalent or higher rating;
 - b) Conform to prevailing POWERGRID specifications;
 - c) Meet applicable BIS, IEC and CEA standards;
 - d) Be procured from approved manufacturers/vendors;
 - e) Be accepted only after inspection and testing by POWERGRID.

POWERGRID **would** have the right to determine technical acceptability of the replacement equipment.

Any issue including diversion, replenishment, return of Regional Spares, or technical acceptance of replacement equipment shall be resolved through the concerned RPC and its sub-forum, as applicable.

6. General Conditions of Diversion

6.1. Timeframe for Diversion and Return

- i. The Borrower should take the physical handover of the equipment from the designated POWERGRID substation within forty-five (45) days from the date of execution of the MoU.
- ii. Failure to take the physical handover of the equipment **within the ninety-days (90) days** from RPC or its sub-forum approval, as applicable, may be treated as revocation the diversion approval.
- iii. **The maximum diversion period for a Regional Spare Transformer or Reactor shall normally be 30 months from the Zero Date of Diversion with approval of RPC. The Borrower may seek an extension by submitting a reasoned request to the concerned RPC. Based on the technical justification and merits of the case, the concerned RPC may approve an extension for a period not exceeding 12 months.**
- iv. The Borrower may ensure return or replenishment of the diverted equipment within this period as per the action plan approved by the RPC or its sub-forum, as applicable, and the provisions of the MoU.

- v. Any request for extension of the diversion period beyond thirty (30) months or delay in taking the physical handover of the equipment may require prior approval of the respective RPC or its sub-forum, as applicable, supported by written justification.
- vi. On completion of the diversion period, the Borrower should return the equipment in healthy condition to POWERGRID to POWERGRID's technical specifications.
- vii. *The "Zero Date of Diversion" shall be the date recorded in the joint handover certificate signed by representatives of POWERGRID and the Borrower at the time of physical transfer from the designated POWERGRID substation or storage location.*
- viii. *The "Date of Return" shall be the date recorded in the joint handover certificate signed by representatives of POWERGRID and the Borrower after successful healthiness checking, at the time of physical transfer at the designated POWERGRID substation or storage location.*
- ix. Force Majeure shall not absolve the Borrower of its obligation to restore, replace, or return the diverted equipment. The equipment shall be returned to POWERGRID in good working condition in accordance with this SOP and POWERGRID's applicable technical specifications.

6.2. Early Recall

- i. RPC Forum may recall any diverted equipment at any time before the expiry of the approved period if required in the interest of grid security or system reliability.
- ii. The Borrower shall return the equipment **within 90 days** upon receipt of such recall notice, and RPC Secretariat shall be intimated accordingly.

6.3. Financial Provisions: Diversion of Regional Spares may be carried out on a cost-neutral basis, ensuring that POWERGRID neither incurs financial loss nor earns additional revenue on account of such diversion. The diversion should not result in any change in the Yearly Transmission Charges recoverable by POWERGRID as per the approved tariff against the diverted asset and financial adjustments, if any, may be settled within the regional pool mechanism as approved by the RPC forum. The Borrower may bear all direct and incidental costs, including transportation, loading/unloading, insurance, erection, testing, and commissioning charges. The Borrower, may be fully liable for any loss, damage, theft, or deterioration of the equipment during the diversion period. Insurance coverage may be obtained in the joint name of POWERGRID and the Borrower with POWERGRID designated as the primary beneficiary.

In line with these principles, the financial treatment for diversion may be as follows:

- i. **ISTS Substations excluding Private Transmission Service Providers (TSPs):** Diversion to any ISTS substation, should be without financial liability to Powergrid/DIC. Such diversion should not affect the regional tariff or cost-sharing framework.
- ii. **STU Substations and Private Transmission Service Providers (TSPs) :** For diversions to STU substations/ Private Transmission Service Providers (TSPs) operating Inter-State Transmission System (ISTS) assets, the equivalent Yearly Transmission Charge (YTC) of

the diverted asset on pro-rate basis, along with any applicable penalty, may be credited to the Regional ISTS Pool for the diverted period.

For this purpose, POWERGRID may raise bilateral bills to the borrower to recover the equivalent YTC of the diverted asset on pro-rate basis, along with any applicable penalty. The YTC may be computed in accordance with the prevailing Tariff Regulations, Sharing Regulations, or any other rules or regulations notified by CERC or the Ministry of Power, and all provisions relating to due dates, late payment surcharge, interest, or other billing conditions may likewise be governed by the applicable rules or regulations. The amount so recovered may be adjusted in the Regional ISTS Pool through the Second Bill mechanism under sharing regulations, and the corresponding credit shall be passed on to the DICs of the Region in the same Second Bill.

6.4. Bank Guarantee (BG) requirements:

- i. Borrower may furnish a valid Bank Guarantee equal to the prevailing cost of the equipment, effective **till 45 days** after return or replenishment is completed.
- ii. The BG may be invoked by POWERGRID to recover any financial loss or liability arising from events such as failure to return or replenish the equipment within the agreed timeframe, or failure to repair, refurbish, or replace the equipment in the event of damage, failure, or loss during transit, erection, or operation.

6.5. Penalty and Default:

- i. If the Borrower fails to return or replenish the diverted equipment within the agreed timeframe, which may **not exceed 30 months and additional extension of 12 months as approved by RPC, if any, a penalty of 15%** of the approved Yearly Transmission Charge (YTC) of the diverted asset may be levied on a pro-rata basis for the delayed duration.
- ii. The penalty amount may be credited to the Regional Component of ISTS charges for the corresponding period. In case of continued default or non-replenishment, POWERGRID may report the matter to the RPC forum for further decision, which may include encashment of the Bank Guarantee and/or regulatory intervention.

6.6. Record-Keeping and Monitoring: POWERGRID may maintain a centralized and up-to-date register of all diversions **and upload periodically on the RPC website**, containing the following information:

- i. Borrower entity name and category.
- ii. Location of Installation by the Borrower**
- iii. Equipment details (type, rating, make, serial number).
- iv. Date of diversion and expected return date.
- v. Physical and test condition at dispatch and upon return.
- vi. Financial treatment and applicable YTC adjustments.

vii. Status of replenishment or replacement.

7. **Review and Amendment:** These SOP may be reviewed from time to time to incorporate operational experience, regulatory updates, or changes in system requirements.
8. **Power to Relax:** RPC forum may relax any of the provisions of these SOP on its own or on an application made before it by the affected party.

Annex B.2.2.a

Agenda of PMJTL :- Insulator Failure

Date: 19.08.2026

Plan for replacement of defective CLR insulator in PMJTL elements

- Status of CLR Procurement **as on 19.08.2026:-**

CLR Rating	Qty under Procurement	Start of progressive supply	Tentative date of end of progressive supply
400kV, 120kN	1682	Jun-26 (Supply started, 6900 Nos. already supplied) 160kN-6200 120kN-700	Oct-26
400kV, 160kN	8591		
400kV Total	10273		
765kV, 160kN	3357	Jun-26 (Supply started, 7520 Nos. already supplied) 210kN-5970 160kN-1550Nos	Nov-26
765kV, 210kN	14428		
765kV Total	17785		

Plan for replacement of defective CLR insulator in PMJTL elements

- Replacement Plan of defective CLR as on 19.08.2026:-

SL No	Name of Line	Total Defective Insulator	Replaced as on Date	Date/Tentative date of start of Replacement work	Tentative No of days required	Tentative date of end of Replacement work
1	400kV New Jeerat-Subhasgram Ckt 1	3382	2900	Jul-26	5	Sep-26
2	400kV D/C New Jeerat-Subhasgram Ckt 2	3382	0	Sep-26	18	Oct-26
3	400kV New Jeerat-Jeerat Ckt 1	1123	0	Sep-26	8	Oct-26
4	400kV New Jeerat-Jeerat Ckt 2	1122	0	Sep-26	8	Oct-26
5	400kV Medinipur-Chanditala Ckt 1	303	0	Oct-26	5	Oct-26
6	400kV Medinipur-Chanditala Ckt 2	303	0	Oct-26	5	Oct-26
7	400kV Medinipur-Kharagpur Ckt 1	329	0	Oct-26	5	Oct-26
8	400kV Medinipur-Kharagpur Ckt 2	329	0	Oct-26	5	Oct-26
9	765kV Ranchi-Medinipur Ckt 1	2530	1500	Aug-26	7	Oct-26
10	765kV Ranchi-Medinipur Ckt 2	2530	452	Aug-26	12	Nov-26
11	765kV Medinipur-New Jeerat Ckt 1	6362	1200	July-26	20	Dec-26
12	765kV Medinipur-New Jeerat Ckt 2	6363	1000	July-26	20	Dec-26
TOTAL		28,058	6600			

Annex B.2.2.c



19 Aug 2026 at 1:13:08 PM

Patna 803311

Bihar

India

Barh



20 Aug 2026 at 11:18:26 AM

Patna 803311

Bihar

India

Barh



Tower Erection work at pile of 400kV Barh - Motihari TL

#Pravin Lal

Patna 803311

Bihar

India

Barh

N 25° 28' 13", E 85° 39' 43"

19 Aug 2026 at 11:33:53 AM



[illegible]

Annex B.2.14

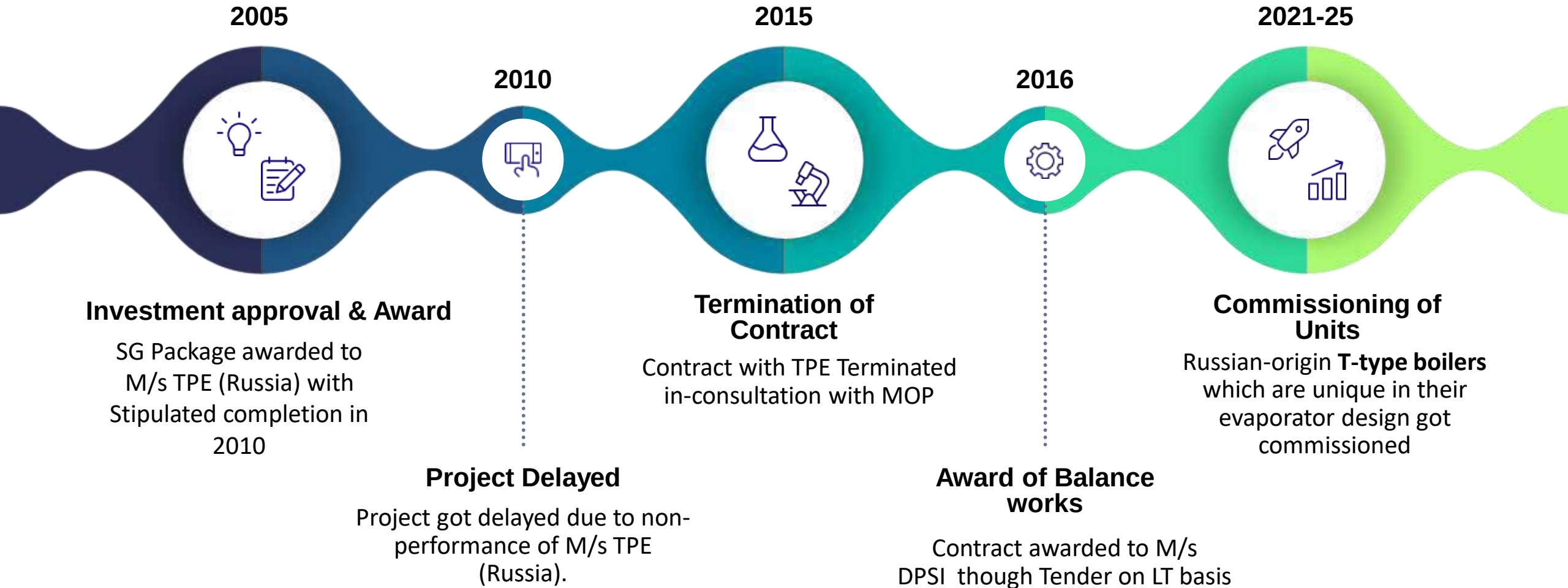
Presentation on BARH Stage#1 (3x660MW) issues





Barh stage-1 timeline

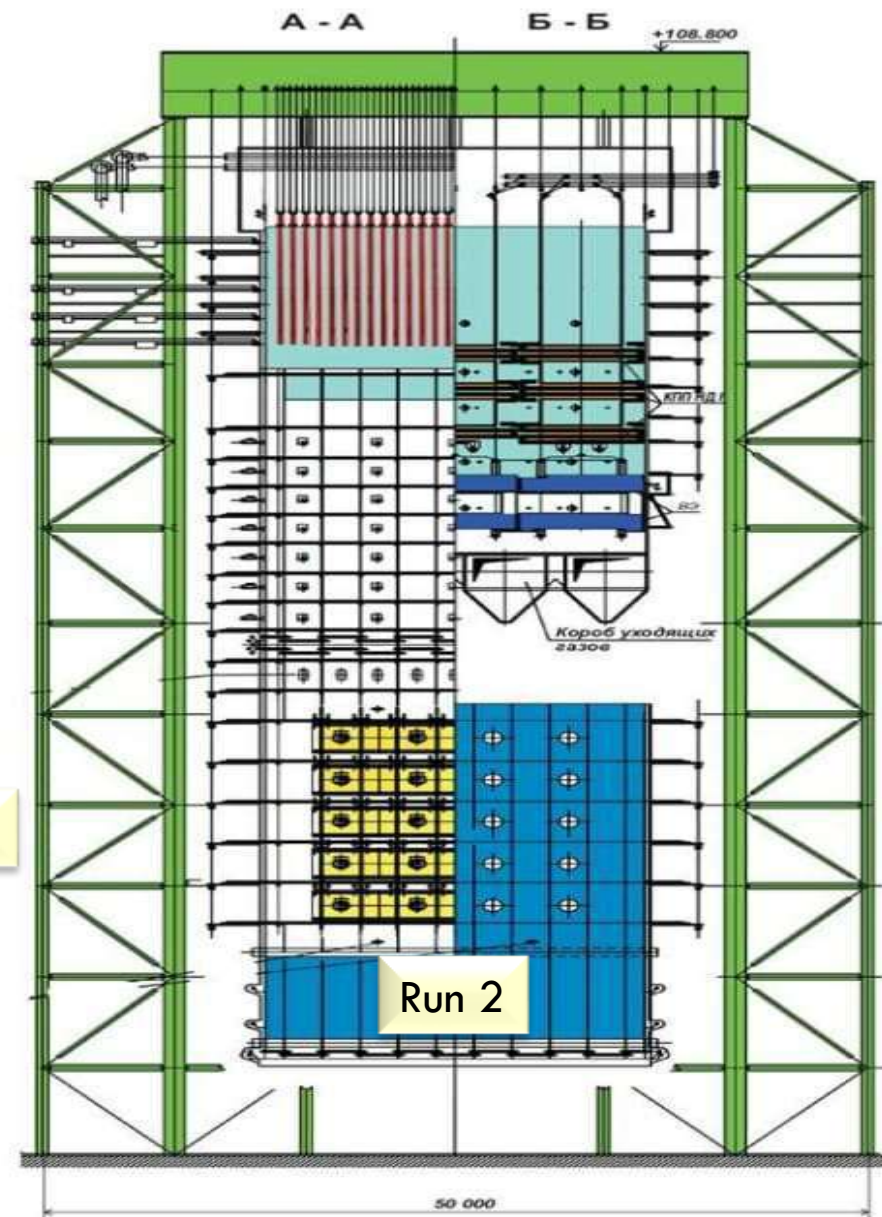
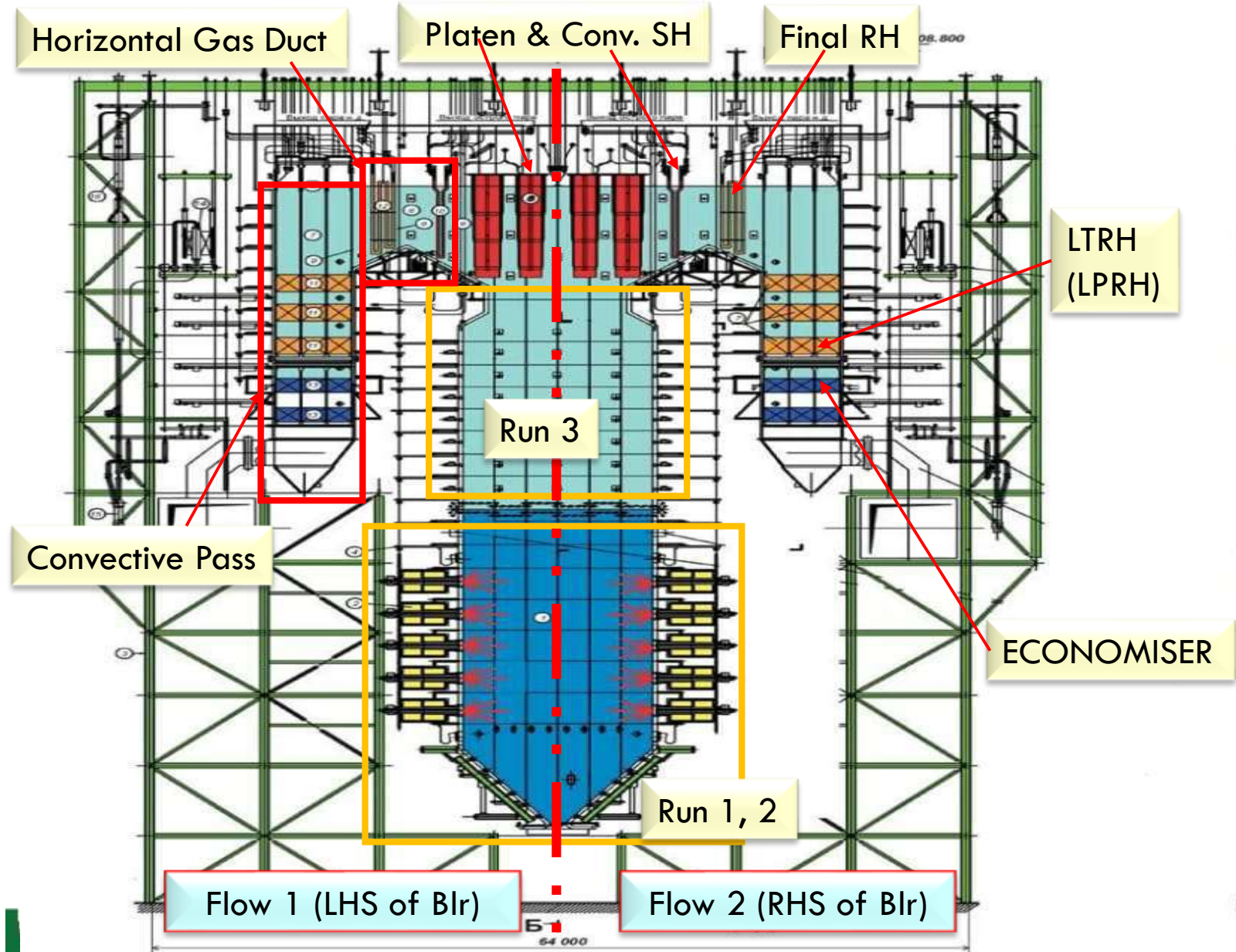
Barh Stage#1(3x660MW) timeline





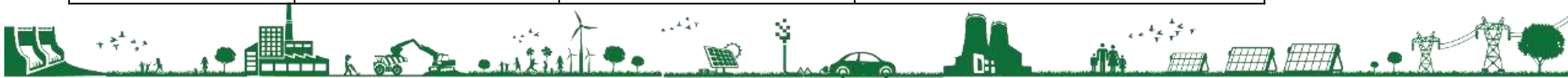
Design arrangement

Barh Stage#1 Boiler arrangement



Barh St-1 Design difference

Description	Barh Stage-I Boiler (M/s TPE)	Barh Stage-II Boiler (M/s BHEL)	Remarks
Boiler Type	T-type boiler with plain bore vertical tubes, multiple header, multiple drain valve, three stages SH spraying etc.	2-pass furnace with spiral and vertical tubes	Plain bore vertical tubes lead to non-uniform heat transfer in the water wall and localized overheating of the tube material at random locations.
Feed water-steam circuit	Two parallel hydraulic flows in boiler extending all the way to superheater outlet steam circuits, each with individual feed control station to control left/right steam temperature balance.	Single FW control with FRS only up to 0-30% range, After that FW control by BFP scoop	Tuning and coordination of individual feedwater flow with combustion is highly challenging due to independent steam-water circuits.
Burners arrangement and combustion air control	Side wall firing with no burner- tilt arrangement, unique SSHE arrangement. Secondary air is controlled by co-centric dampers having electrical actuators for each side individually	Burners installed at 4 corners with burner tilt arrangement making fireball in the centre of furnace Secondary air is controlled by pneumatic dampers with a single command to all 4 corners	The absence of burner tilt arrangement (which helps in controlling furnace heat distribution) inside wall-fired boilers leads to non-uniform heat transfer across water wall tubes. Further, individual combustion air control systems operated through multiple electric actuators leads to unequal air distribution across burner elevations, resulting in localized combustion imbalance and non-uniform furnace heat distribution.





Operational Parameters

Barh Stage#1 Operational parameters

Particular	UoM	FY 22-23	FY 23-24	FY 24-25	FY 25-26
DC	%	81.0	67.7	72.1	72.4
Forced Outage	%	14.77	16.03	17.25	21.43
Forced Outage due to BTL	%	11.85	12.95	14.5	17.56
APC	%	6.80	7.45	7.47	7.72
SOC (Sp. Oil consumption)	ml/kwh	1.50	3.56	3.18	3.59
PLF	%	70.4	60.0	63.7	59.86

This FY till date FO due to BTL:15.78%



No of outages due to BTL



Unit	FY 21-22	FY 22-23	FY23-24	FY24-25	FY25-26	FY 26-27 (till 28.06.26)
U1	8*	11	18	11	15	4
U2	-	-	6*	16	18	4
U3	-	-	-	-	7*	2
Total	8	11	24	27	40	10

U#1 COD: Nov-2021
U#2 COD: Aug-2023
U#3 COD: July-2025



Barh-I comparison with other NTPC supercritical units



FY	DC (%)		FO (%)		BTL (%)	
	Barh-I	NTPC super-critical units (except Barh-I)	Barh-I	NTPC super-critical units (except Barh-I)	Barh-I	NTPC super-critical units (except Barh-I)
2021-22	74.03	86.70	25.79	3.99	20.36	2.49
2022-23	81.01	92.15	14.77	3.61	11.85	2.33
2023-24	67.70	90.10	16.03	3.39	12.95	1.91
2024-25	72.09	89.31	17.05	3.06	14.50	2.14
2025-26	72.37	90.00	21.43	3.51	17.56	2.66





Experts help/CEA relief
sought

Communications done with OEMs/Experts/CEA

S. No	Description	Name of the Vendor/s	Date of meeting /communications	Status
1	Regarding Critical Issues in Stage-I (3x660 MW) Boilers at NTPC BARH	M/s TKZ	12-Mar-24	Non- responsive
2	Regarding Critical Issues in Stage-I (3x660 MW) Boilers at NTPC BARH	M/s BHEL	04-Apr-24	Non- responsive
3	TKZ-Engineering Issues of Stage-1	M/s TKZ	22-Nov-23	No improvement
4	Expression of interest for study and resolving issues in Russian make 660MW Supercritical Boilers in Barh Stage#1	M/s BHEL	14-Aug-24	Non- responsive
5	Expression of interest for study and resolving issues in Russian make 660MW Supercritical Boilers in Barh Stage#1	M/s L&T MHI	14-Aug-24 Vendor mail 28-Nov-24	Regretted to support NTPC
6	Record Notes of Meeting on 26th February 2025 at BHEL Trichy	M/s BHEL	26-Feb-25/10-June-25	Regretted to support NTPC
7	NTPC letter to CEA for relief for base load operation and lower ramping		25-Sep-2024	
8	Study from third party on advice of CEA	M/s Thermax	23-Sep-25 PO awarded for study of Barh Stg-1 issues	Final report Submitted in June-26





Conclusion/Action plan

- The boiler **design of Barh Stage-I units is unconventional and unique**, having T-Type construction, effectively operating as two conventional boilers arranged back-to-back having **two separate steam-water circuits** and separate Feed water regulating system.
- Current observations indicate that these three supercritical units are **unable to operate reliably under low-load conditions below 80%** or during frequent ramp-up and ramp-down cycles. Operating the boiler under such regimes results in significant deviations from design parameters, leading to **premature metallic degradation and frequent tube failures**.
- As an interim measure, **it is recommended to avoid frequent load cycling**. These **units should be operated at a base load of not less than 80% TMCR** under supercritical conditions to enhance reliability and minimize component stress with the present pressure part configuration.
- Given the challenges outlined above, **detailed technical study is essential**. This should include extensive data collection and analysis to develop a sustainable solution. Potential areas of improvement may include revised operational strategies, control logic modifications, combustion tuning, loop tuning, physical design changes, etc.
- **The entire process is anticipated to take approximately three years which includes** Study by Boiler OEMs/independent experts, evaluation of recommendations with finalization of modification action plan and resource mobilisation, followed by implementation in one unit with performance assessment and subsequent rollout to remaining two units.





भारत सरकार
विद्युत मंत्रालय
केंद्रीय विद्युत प्राधिकरण
सचिव कार्यालय

Phone:11-26732203; E-mail: secretary.cea@nic.in

दिनांक: 13.08.2026

Head of Project,
NTPC Barh STPS
Barh, Patna, Bihar
Postal Code:803215

Subject: NTPC request seeking relaxation from Central Electricity Authority (Flexible Operation of Coal based Thermal Power Generating Units) Regulations, 2023 in terms of Minimum Power Level and Ramp Rates for Barh-I STPS (3x660 MW) due to technical and operational issues -reg.

Dear Sir,

Reference is invited to NTPC letter dated 01.06.2026 and other previous correspondence on the above-mentioned subject vide which NTPC requested CEA to provide relaxation for Barh-I STPS (3x660 MW) in terms of Minimum Power Level and Ramp Rates from Central Electricity Authority (Flexible Operation of Coal based Thermal Power Generating Units) Regulations, 2023.

In this regard, the proposal was discussed in the Authority Meeting held on 27.07.2026 and decision of the Authority is attached at Annexure.

Encl: As above

सचिव
13.8.26
सचिव, के. वि. प्रा.

प्रतिलिपि:

1. चेयरमैन एवं मैनेजिंग डायरेक्टर, एन.टी.पी.सी. लिमिटेड
2. निदेशक (संचालन), एन.टी.पी.सी. लिमिटेड

सूचनार्थ प्रतिलिपि:

1. अध्यक्ष, केंद्रीय विद्युत प्राधिकरण
2. सदस्य (तापीय/ हाइड्रो/ पावर सिस्टम/ प्लानिंग/ ई एंड सी / जीओ एंड डी), केंद्रीय विद्युत प्राधिकरण

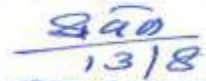
सचिव/Secretary
केंद्रीय विद्युत प्राधिकरण/C.E.A.
विद्युत मंत्रालय/Ministry of Power
भारत सरकार/Govt. of India
नई दिल्ली/New Delhi-66



Annexure

NTPC request seeking relaxation from Central Electricity Authority (Flexible Operation of Coal based Thermal Power Generating Units) Regulations, 2023 in terms of Minimum Power Level and Ramp Rates for Barh-I STPS (3x660 MW) due to technical and operational issues -reg.

1. In view of the **exceptional circumstances** and unique boiler design, the Authority agreed to give **relaxation to Barh-I STPS for three (03) years** (from the date of issue of MoM of Authority Meeting i.e. 12.08.2026) from Central Electricity Authority (Flexible Operation of Coal based Thermal Power Generating Units) Regulations, 2023 as per **regulation 8** of the said regulations, as mentioned below:
 - i. **Relaxation for Minimum power level to 80%** - [Regulation 6 (Minimum power level capabilities of coal based thermal power generating units for flexible operation)]
 - ii. **Relaxation in the ramp rate to 0.1 %** of capacity corresponding to MCR per minute [Regulation 7 (Ramp rates capabilities of coal based thermal power generating units for flexible operation)]
2. Authority explicitly mentioned that relaxation given to Barh-I STPS from Central Electricity Authority (Flexible Operation of Coal based Thermal Power Generating Units) Regulations, 2023 is only a technical relaxation and for all commercial issues arising out from this relaxation, NTPC has to approach CERC.
3. Authority also directed NTPC that award process for study, recommendations and suggestions given by M/s Thermax may be implemented within stipulated time of three (03) years. Further, EPRI recommendation should also be completed in parallel to the Thermax recommendation.
4. Authority directed that an in-house team of NTPC and/or any other experts may be explored by NTPC so that resolving the issue do not exceed beyond a period of 3 years. Authority also suggested NTPC to maintain the fuel quality at plant to reduce the outages in the units for reliable operation of Barh-I STPS units.
5. Also, NTPC shall submit the quarterly reports to CEA regarding the progress of implementation of recommendation/modifications/rectifications/suggestions given by various experts.


 13/8
 सचिव/Secretary
 केन्द्रीय विद्युत प्राधिकरण/C.E.A.
 विद्युत मंत्रालय/Ministry of Power
 भारत सरकार/Govt. of India
 नई दिल्ली/New Delhi-66



Thank You



Website: www.ntpc.co.in